

Title: Electrofluidic Fiber Muscles

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Abstract:

A fiber form-factor makes biological muscles highly modular, scalable, and well-integrated in animals' bodies, constituting 50% of human body weight. The servo motors that drive today's robots lack the flexibility and modularity of muscle fibers, limiting integration and dexterity.

Here, we report electrofluidic fiber muscles, a class of soft artificial muscles for robots and wearables with power density comparable to skeletal muscles (50 W/kg), contraction strains of 20%, and response time of 0.3 seconds. These 2 mm thick muscles are made of antagonistic thin McKibben pairs driven by electrohydrodynamic fiber pumps in a closed circuit. They require no external reservoir, are electrically driven, untethered, and silent.

We modeled their coupled dynamics and discovered that performance is dramatically increased by pre-pressurizing the muscle at an optimal bias pressure. Bias pressure allows the antagonist actuator to act as a reservoir for the agonist, enables 200% higher voltages by preventing cavitation, and leverages the nonlinear response of McKibben actuators, increasing strain 3x at a given pump pressure. We characterized muscle strain and response time, showing that optimal bias pressures exist. With unprecedented versatility and modularity, electrofluidic muscles scale by simply bundling fibers. By selecting the ratio between pumps and actuators, we programmed their performance for different robotic tasks: a fast lever (180 mm/s) that throws objects in <0.3 seconds; a strong bundle that lifts 4 kg (200x its weight) with 30 mm stroke; a woven muscle that bends a robot arm by 40° in <2 seconds and compliant enough for a human handshake.

One-Sentence Summary: Bias pressure is key to making modular and versatile electrofluidic fiber muscles with power density comparable to mammalian skeletal muscles

Main Text:

INTRODUCTION

Nearly all animal muscles are fiber format, while nearly all robots are driven by electromagnetic (EM) servo motors, with bulky cylindrical shapes. EM motors offer high power density, are efficient and reliable, yet their form factor limits integration in robots with soft body parts (1) and in wearable robots (2). Transmitting motion to anything that is not a rotary joint requires additional components such as pulleys and tendons. On the contrary, biological muscle fibers can be bundled in different formats, with simple force scalability by increasing the number of fiber muscles in each bundle.

To overcome the limitations of EM motors, researchers developed fluidically and electrically driven soft actuators. Easy to manufacture, robust, and scalable, soft fluidic actuators have been demonstrated driving soft grippers (3–6), walking robots (7, 8), swimming robots (9–11), wearables (12, 13) and even large-scale deployable robots (14). Their main limitation is the bulky pneumatic infrastructure required to operate them, which has so far hindered applications, except for industrial grippers. Conversely, electrically driven actuators have the advantage of directly converting electrical energy from a battery into mechanical work. Dielectric elastomer actuators (DEAs) have been the gold standard for years (15–19): soft, fast, and high in power density. However, these actuators are difficult to scale up and their membrane form factor complicates integration in robots. Recent electrostatic and electrohydraulic actuators (20–26) overcome some of these limitations, gaining robustness and scalability, while preserving high power density. These artificial muscles have flat or 3D form factors, and their versatility and modularity have limitations: each class of actuators works well in specific embodiments, but major re-engineering is required when one wants to develop a different application (27).

Fiber-format, electrically driven soft actuators exist (table S1). DEA fibers (28, 29) are compact and monolithic, yet response time and power density remain below biological muscles (50 W/kg). Shape memory alloys (30–33) tend to have high force density (40 N/g) but slow return times due to cooling time constants (> 10 s). They can reach high temperatures ($\geq 50^\circ\text{C}$) and are made of stiff metals, making the interface with soft materials challenging. Other polymer-based electrothermal actuators show much lower power density and require high temperatures (200°C) (34, 35). CNT-based twisted polymer actuators reach 50 W/kg power density only if operated in an electrolyte bath (36). Modular electrically-driven actuators with < 20 ms response time and high power density (122 W/kg peak, 57 W/kg average) have recently been introduced, though with a prismatic form factor and a rigid exoskeleton as part of their body (37). There is still a need for highly modular and reconfigurable artificial muscles in soft fiber format with power density ≥ 50 W/kg. Such soft actuators are a key enabler of multi-functional robots with flexible bodies and high-dexterity soft wearable robots.

In this work, we report fiber-format electrofluidic artificial muscles with power density comparable to mammalian muscles (50 W/kg average) that are modular, versatile and reconfigurable. They are made of antagonistic thin McKibben fluidic actuators driven by ElectroHydroDynamic (EHD) fiber pumps, connected in a closed system (Fig. 1A, Movie 1 and movie S1). The electrofluidic fiber muscles have 1 - 2 mm diameter, contraction strains near 20% with response times < 0.3 s. They are directly driven by an electrical input and operate without noise. The minimal configuration is a muscle pair unit: one fiber pump connected between an antagonistic pair of McKibben actuators (Fig. 1B).

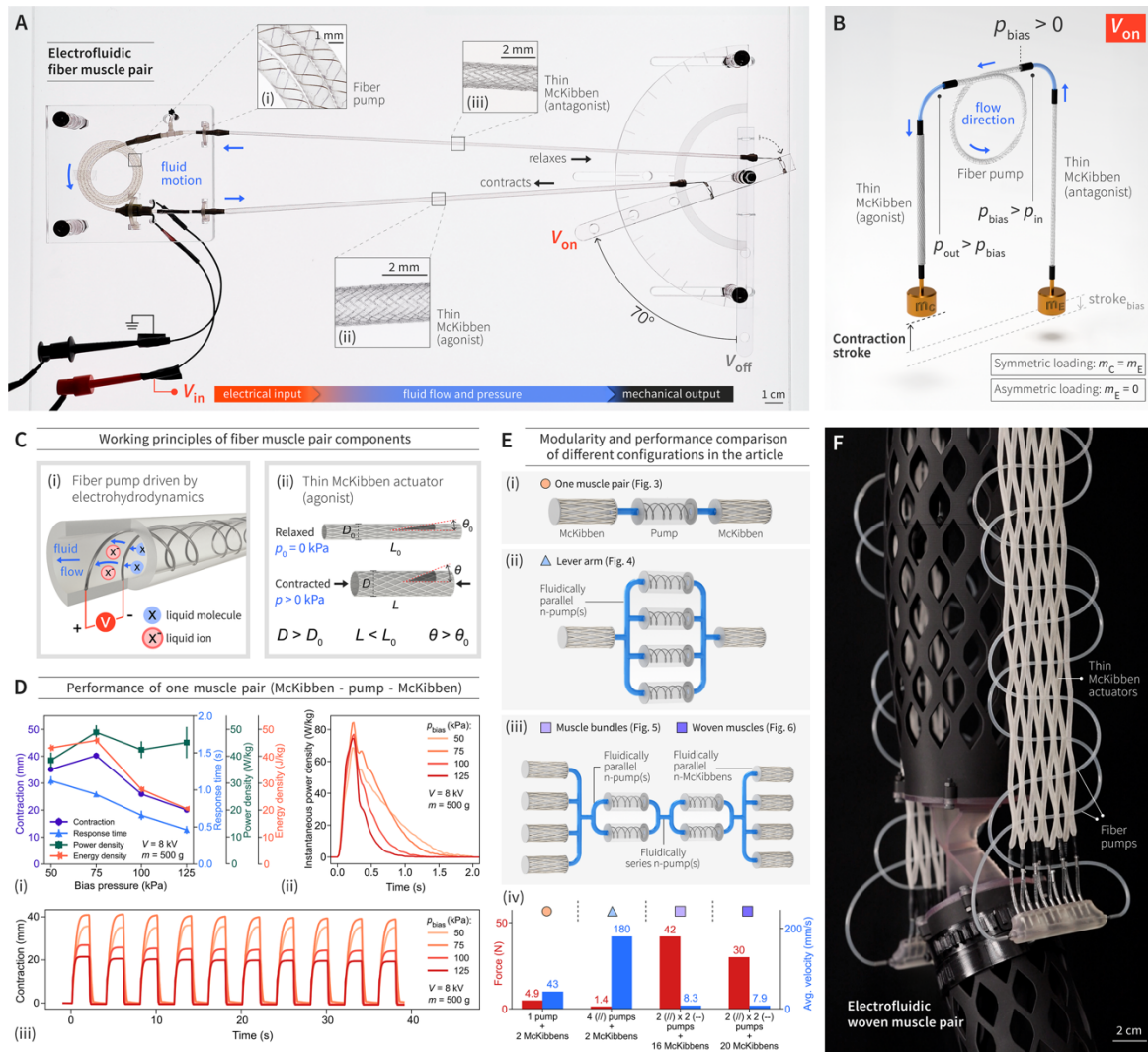


Fig. 1. All-fiber electrofluidic artificial muscles with large design freedom. (A) Each electrofluidic fiber muscle pair is composed of two antagonistic thin McKibben actuators (ii, iii) connected by an EHD fiber pump (i), each < 2 mm diameter. (B) Muscle pair unit working principle: one fiber pump transfers fluid between an antagonistic pair of McKibbens. Pre-filling the system with extra liquid causes both muscles to pre-contraction. When voltage is applied, the antagonist extends, while the agonist contracts. Each muscle is connected to a load. Symmetric loading: masses m_C and m_E are equal. Asymmetric loading: $m_E = 0$. (C) The muscle converts electrical input into fluid pressure, inside the fiber pump (i), and fluid pressure into mechanical output in the McKibben actuators (ii). (D) A single muscle pair has an average power density of ≥ 50 W/kg (asymmetric loading), comparable to skeletal muscles (i). Error bars show $\pm$ SD for $n = 3$ devices. Instantaneous power density (ii) and contraction (iii) over time vs bias pressure. (E) McKibbens and pumps can be modularly arranged for different robotic tasks. The simplest electrofluidic fiber muscle (i) consists of one pump and two McKibbens (one muscle pair). Multiple pumps in parallel increase velocity (ii), see lever arm demo. Scaling up both pumps and McKibbens number increases force (iii), see Muscle bundles and Woven muscles demoes. Average force and velocity comparison of different modular configurations in the article (iv). (F) Modular scalability allows architected muscles for robotics, including compliant woven muscle pairs on robotic arms.

Thin McKibben actuators (Fig. 1C ii) are power-dense, compliant, and have small diameters (< 2 mm) (38). Their fiber form factor makes them a highly versatile alternative to electromagnetic motors for applications where integration is key, such as wearable exosuits and dexterous robots. To date, McKibben actuators required external pumps and valves, with few exceptions (39), limiting untethered operations. Fiber pumps (Fig. 1C i) are mm-sized flexible tubes that generate their own pressure and flow rate (40). Soft and solid-state,

these pumps solve the long-standing problem of integrating pressure generation in portable systems. Our electrofluidic muscles are pre-filled with a dielectric fluid and the fiber pumps move it in and out of each McKibben actuator, which contracts or extends in response. As a result, these all-fiber-format muscles convert electrical energy from a power supply into fluid pressure in the pumps, and then fluid pressure into mechanical work in thin McKibben actuators (Fig. 1A, Movie 1). The result is an electrofluidic fiber muscle that contracts/extends in response to voltage, whose shape and performance can be designed by changing the combination of its two components (McKibben actuators and fiber pumps).

We discovered that the performance of these muscles is dramatically influenced by the bias pressure applied when pre-filling them with a dielectric fluid. The electric field that the closed system tolerates when a bias pressure of 75 kPa is applied is 200% of that without bias (limited by fluid cavitation), resulting in a dramatic 7-fold increase in the contraction stroke with the same load (14 % vs 2% at 2 N force). Also, bias pressure enables leveraging the intrinsically nonlinear response of McKibben actuators, further increasing the stroke at fixed voltage. With 5 N load, a bias pressure of 75 kPa brings the contraction strain from less than 2% (no bias) to over 6% (3-fold increase) at an applied voltage of 6 kV. At the optimal bias pressure value, our Electrofluidic muscles reach performance comparable to skeletal muscles (Fig. 1D).

We report a series of results to support our claim that bias pressure is a key requirement to operate fluidic actuators with an integrated pump in closed system. First, we characterized the performance of the individual muscle pairs (one fiber pump with two McKibben actuators connected in an antagonistic configuration) in terms of force, stroke and response time. We measured the influence of bias pressure on performance and on the maximum electrical voltage that can be applied to the system. We also developed a model that describes the dynamic coupling of EHD fiber pumps with thin McKibben actuators. This model can be used as a design tool for electrofluidic muscles in robotic tasks. Given the force and response time required by the task, the model outputs the number and size of the fluidic actuators, the number of fiber pumps and their series-parallel arrangement. We characterized the electrofluidic muscles dynamically and report their stable operation over 5000 cycles. Finally, the high versatility and modularity is supported by the three robotic embodiments that we built and characterized.

Featuring a versatility unprecedented in soft actuators and even in biological muscles, the performance of electrofluidic fiber muscles can be programmed by design, simply selecting the number of McKibben actuators to be bundled together and the number of fiber pumps to connect in series and/or in parallel (Fig. 1E). More actuators in parallel lead to higher forces. A higher ratio of fiber pumps in parallel over fluidic actuators produces a faster response. We show that it is possible to harness such versatility and large power density by building three different robotic demonstrators using the same actuator-pump-actuator ensemble (Fig. 1E): 1) a fast lever (180 mm/s) that can throw multiple objects in the air in < 0.3 seconds, 2) a strong muscle bundle that can lift > 4 kg (200x its own weight) with 30 mm of stroke (14 % strain), 3) a flat woven muscle (Fig. 1F) that bends a robot arm by over 40 degrees in < 2 seconds and compliant enough that a human can shake its hand.

With their fiber form factor, high power density and unique modularity, electrofluidic fiber muscles can fill the need for scalable and versatile fiber-shaped artificial muscles for soft and wearable robotics (2, 41).

RESULTS

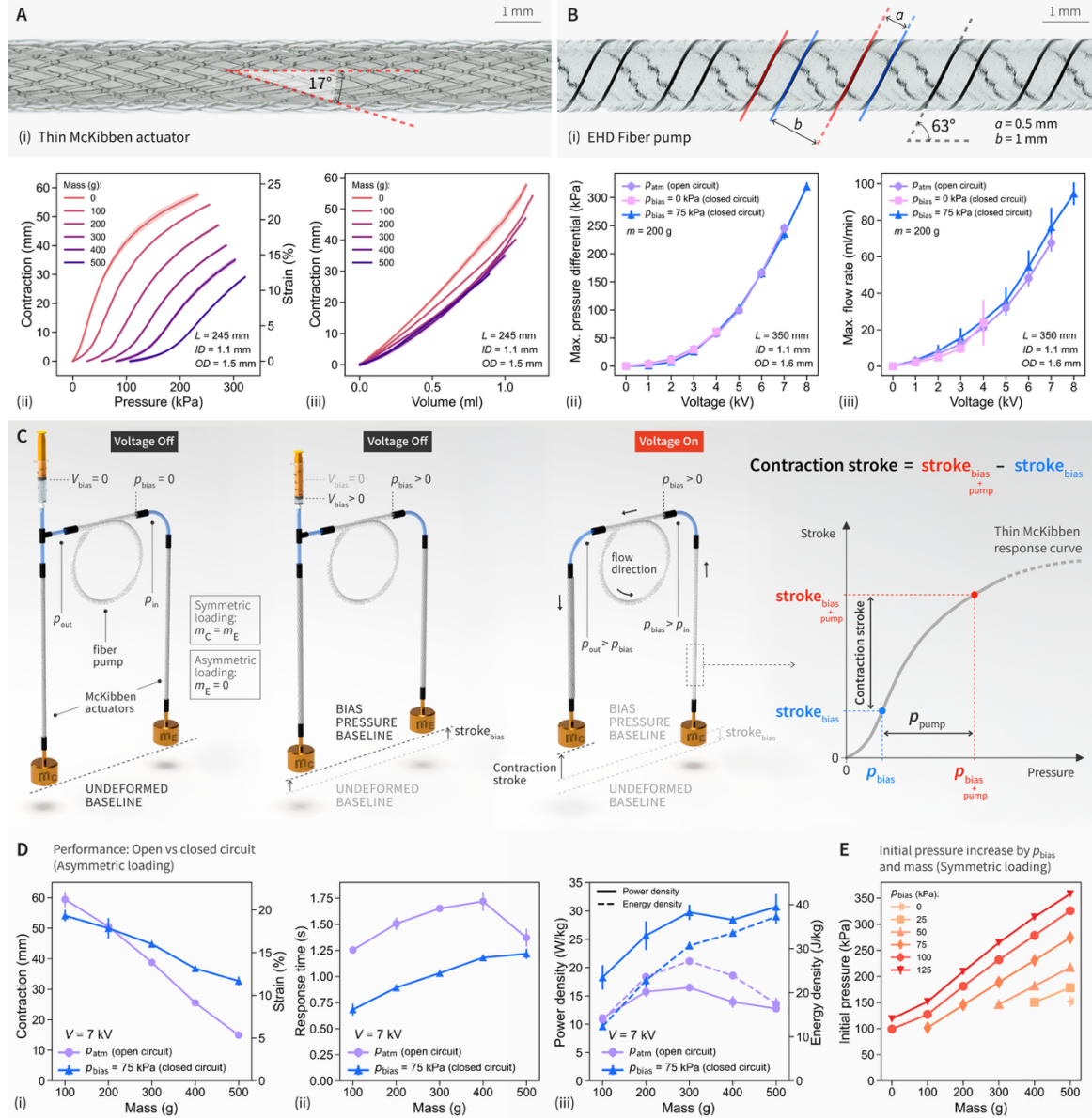


Fig. 2. Electrofluidic fiber muscles as coupled systems composed of thin McKibben actuators and fiber pumps. (A) Characterization of thin McKibben actuators ($ID = 1.1$ mm, $OD = 1.5$ mm, $L = 245$ mm) showing nonlinear response (i). Contraction as a function of pressure (ii) and volume (iii). (B) Characterization of fiber pumps ($ID = 1.1$ mm, $L = 350$ mm) (i). Maximum pressure differential (ii) and maximum flow rate (iii) generated by the fiber pump at different applied voltages under three different fluidic circuit configurations. The performance at the same voltage is unchanged, but the closed circuit at 0 kPa bias pressure shows early breakdown above 4 kV due to cavitation. (C) Two antagonistic McKibbens are coupled with one fiber pump transferring liquid between them. By regulating the bias pressure, we leverage the nonlinear response of McKibbens, improving the contraction stroke and response time for a given pressure and flow rate that the pump generates. (D) Applying bias pressure (p_{bias}) increases the muscle performance compared to same muscle operated as an open system at atmospheric pressure (p_{atm}) with an external liquid reservoir, especially at higher loads. Contraction (i), response time (ii), power density and energy density (iii) as a function of the mass attached to the contracting McKibben (m_c), at 7 kV. (E) Initial pressure of the system changes for given values of bias pressure and because of hanging mass to both muscles (symmetric loading). Shaded area/error bars represent $\pm$ SD over $n = 3$ devices.

Closed circuit configuration of Electrofluidic Fiber Muscles for out of the lab robotic applications and importance of bias pressure

To study Electrofluidic fiber muscles as a system, we first characterized its two components, thin McKibben actuators (Fig. 2A) and fiber pumps (Fig. 2B). Details on design, materials and characterization experiments are in Materials and Methods and in Supplementary Materials and Methods.

We aim to develop artificial muscles for dexterous robots and wearable soft exosuits. These applications require a degree of portability and integration that are hardly compatible with a reservoir open to atmosphere. Evaporation of the dielectric liquid, the need to maintain vertical reservoir orientation to prevent liquid leakage, and multiple reservoirs required for multiple muscles are key limitations of operating electrofluidic muscles in an open circuit configuration. Therefore, we designed our artificial muscles for closed-circuit operation.

This transition from open to closed circuit introduced two hard challenges: (1) storing extra liquid required to fill the McKibben actuators during operation, and (2) cavitation near the pump inlet when local pressure drops below the dielectric liquid's vapor pressure. We addressed both challenges by designing antagonistic muscle pairs and filling the system with extra amount of liquid, until reaching a bias pressure above atmospheric pressure.

The configuration and filling procedure are shown in Fig. 2C. Two McKibben actuators are connected to either side of one fiber pump. The system is filled with liquid and an extra amount is added, making both actuators contract by a $\text{stroke}_{\text{bias}}$, determined by the amount of extra liquid and resulting bias pressure (p_{bias}) reached when filling. The electrofluidic muscles are operated by applying a DC voltage to the fiber pump. When voltage is off, the whole system (pump and two actuators) is at constant pressure, equal to p_{bias} . When voltage is turned on, the pump moves liquid from one actuator to the other. The antagonistic configuration ensures one McKibben actuator (the extending one) acts as a reservoir for the other (the contracting one) and vice versa. The pressure in the contracting actuator (pump outlet side, p_{out}) increases ($p_{\text{out}} > p_{\text{bias}}$), while the pressure in the extending one (pump inlet side, p_{in}) decreases ($p_{\text{in}} < p_{\text{bias}}$) until the pressure difference ($p_{\text{out}} - p_{\text{in}}$) equals the maximum pressure differential that the pump can generate at an applied voltage. At this point, the system reaches steady state. The contracted muscle lifted its connected load to a contraction stroke (s_C) above the bias pressure baseline, producing mechanical work output. The extending actuator extends of a different stroke, always $\leq \text{stroke}_{\text{bias}}$. When voltage is turned off, pressure difference drives the fluid backwards and both actuators return to the bias pressure baseline. Contraction (s_C) refers to the contraction stroke of the agonist McKibben, from the pre-pressurized state (bias pressure baseline) to the actuated state. Elongation (s_E) refers to the extension stroke of the antagonist McKibben, from the pre-pressurized state (bias pressure baseline) to the actuated state.

We tested two loading conditions: symmetric (equal loads on both sides of the muscle pair $m_C = m_E$) and asymmetric (contracting McKibben loaded and extending one unloaded $m_E = 0$) (Fig. 2C). The asymmetric loading condition is the one we always use to compute the energy and power density of the system, since in this condition it is clear that all the mechanical work on the lifted load (m_C), connected to the contracting actuator, is generated by the electrofluidic muscle pair during its contraction phase.

Using an optimal bias pressure (p_{bias}) when filling the system has four important implications: (1) enables higher pressure operation of the system (higher voltage can be

applied to the fiber pumps) by preventing cavitation at the pump inlet, which would disrupt operation; (2) it leverages the nonlinear response of McKibben actuators, optimizing pump-actuators matching and increasing power density output of the system; (3) it allows the extension of the antagonist muscle, increasing joints range of motion in flexion-extension movements in musculoskeletal robots (44); (4) it makes McKibben actuators back stretchable, a desirable feature for compliant interaction with humans and the environment. Cavitation refers to the local boiling of a liquid (typically at the pump inlet) when the local pressure drops below the liquid's vapor pressure. It is a disruptive phenomenon for fiber pumps since it triggers electrical breakdown (breakdown strength of a vapor is always dramatically lower than the one of the corresponding liquid). For the Novec 7100 dielectric liquid used in this work, the vapor pressure is 27 kPa (absolute pressure) at 25 °C (45). Local pressure of the liquid in the pumps is a sum of static pressure (measured by our pressure sensors) and dynamic pressure, due to liquid acceleration, which is < 0 near the emitting electrode. As a result, cavitation can occur for static pressure significantly higher than the vapor pressure of the liquid. Movie S2 shows how cavitation occurs in our system when static pressure drops to a value close to the dielectric liquid's vapor pressure. We discovered that cavitation is prevented by pre-filling with extra liquid such that bias pressure p_{bias} ensures that pressure at the inlet (p_{in}) stays always $>$ vapor pressure of the dielectric liquid (with a sufficient safety margin).

We also report how filling the system at an optimal bias pressure improves its performance both in terms of stroke (operating under load) and response time, leveraging the nonlinear response of McKibben actuators (Fig. 2C, right). While bias pressure reduces the total maximum contraction stroke that can be achieved for a given McKibben actuator (since the muscles are pre-contracted by bias pressure), the contraction stroke per given finite pressure generated by the fiber pump is higher, since the fiber pump pressure (p_{pump}) is applied in the high slope region of the McKibben's stroke-pressure response curve. A similar effect also applies to the stroke-volume response curve of the actuators (Fig. 2A, ii and iii).

We characterized our muscle composed of two McKibben actuators and one fiber pump in both open circuit (p_{atm}) and closed-circuit configurations (with $p_{\text{bias}} = 75$ kPa) and compared their performance (Fig. 2D). Our data show how the contraction stroke for closed circuit overcomes the one for open circuit when the load added to the agonist actuator is > 200 g. The response time for the closed-circuit with bias pressure is always lower than the one for the open circuit. A combination of these two results leads to a power density always higher for the closed-circuit case and the difference increases by increasing the applied load, reaching nearly 250% ratio at a 500 g load (movie S1). Energy density follows the same trend of contraction stroke: the energy density of the closed circuit with bias pressure overcomes the one of the open circuit for loads > 200 g. Computation methods for energy density, power density and other metrics are detailed in Supplementary Materials and Methods.

While performance also depends on the specific fiber pump characteristics, and a longer pump producing higher pressure can lead to higher performance, there are two important considerations: (1) longer pumps come at the cost of increased fabrication complexity and power requirements; (2) our model proves that there is a maximum pressure that can be applied to a given pair of closed-circuit McKibben actuators in order to maximize stroke, since a higher pressure would lead to cavitation, making pump operation impossible (Supplementary Text).

When we add loads to the McKibben actuators, the internal pressure of the closed system increases. We define “initial pressure” as the pressure throughout the system after applying bias pressure and attaching loads. Figure 2E shows measured initial pressures for different values of bias pressure and different loads added to each actuator. This initial pressure determines how much pressure can be applied by the pump before reaching cavitation and/or McKibben actuators burst pressure limit (550 kPa for the actuators used in this work).

In summary, a closed-system design for electrofluidic fiber muscles is required for portability and integration in robotic and wearable applications beyond the lab. We discovered that by filling the system with extra liquid until reaching an optimal bias pressure, the performance can significantly exceed that of the same system operated in open circuit-configuration. The following section presents our physical model and thorough experimental characterization of individual muscle pairs in closed-system configuration, revealing the optimal bias pressure that maximizes performance of a given system.

Physical model of coupled-system dynamics for Electrofluidic Fiber Muscles

Electrofluidic muscles are electro-fluido-mechanical transducers. They transform electrical power (voltage times current $U \times I$) into fluidic power (flow rate times pressure $Q \times p$) inside fiber pumps, and fluidic power into mechanical power (force times velocity $F \times v$) inside McKibben actuators. They work as a coupled system: fiber pumps response influences McKibben actuators and vice versa. Therefore, to capture the behavior of electrofluidic muscles and the influence of the different parameters (geometry, materials, bias pressure), we need a coupled system model (fig. S1).

We first modeled thin McKibben actuators mapping pressure p and volume V inside the actuators to force and displacement, for a given geometry (diameter, length, braid angle) and tubing material. For fiber pumps, we used a linear fit of the experimental response curves, mapping applied voltage to a flow rate – pressure line. Fiber pumps response is practically linear due to low Reynolds number (40), so a linear fit maps it well. A physical model for fiber pumps, accounting for the multi-physics, multi-scale set of mechanisms is still an open research topic and goes beyond the scope of this work. By coupling the fiber pump response curves with the McKibben model, we obtained a coupled system model that predicts the dynamic interaction between these two components under specific load and voltage conditions.

The model reveals how the dynamics of electrofluidic fiber muscles are influenced by the pump and actuator parameters. Thanks to the successful experimental validation, the model can serve as a design tool for predicting behavior of muscle systems than those tested in this article. Moreover, starting from a robotic application with given target force, stroke and response time, the model can predict the required number of McKibben muscles, fiber pumps connected in series and/or parallel to accomplish the task. It also predicts the optimal bias pressure to maximize contraction and response time.

All the details of model derivation and experimental validation are included in Supplementary Text and fig. S1 and S2.

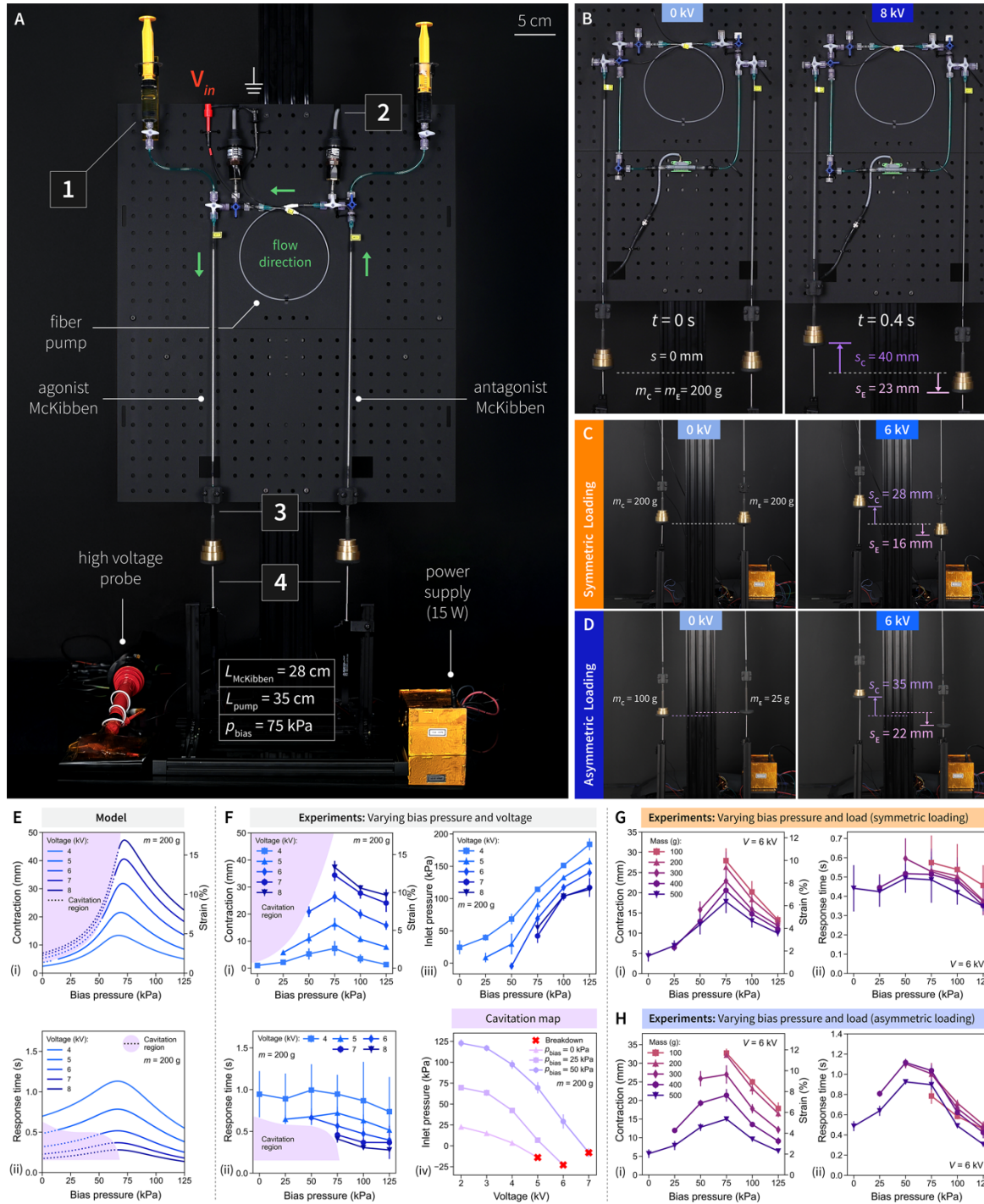


Fig. 3. Characterization of individual muscle pairs. (A) Experimental setup consists of 1) two fluid reservoirs, 2) two gauge pressure sensors, at the pump inlet and outlet, 3) two load holders, and 4) two displacement sensors. (B) Muscle response at 8 kV under 200 g symmetric loading: contraction stroke $s_c = 40 \text{ mm}$ (agonist) and extension stroke $s_e = 23 \text{ mm}$ (antagonist) in 0.4 s. Actuation tests at 6 kV under (C) symmetric loading (200 g on each actuator) and (D) asymmetric loading (100 g on contracting actuator only). (E) Model prediction for contraction (i) and response time (ii), and (F) experimental results of contraction (i), response time (ii) and inlet pressure (iii); for different bias pressures, at voltage values ranging from 4 to 8 kV, under a fixed symmetric load of 200 g. Pressure measured at the pump inlet (iv) that correlates to the cavitation regions shown in (i) and (ii). When gauge pressure decreases below 0, there is a high risk of cavitation and pumps cannot be operated (movie S2). Contraction (i) and response time (ii) at 6 kV, under different bias pressures, and applied mass ranging from 100 – 500 g, under (G) symmetric loading and (H) asymmetric loading. All values with error bars correspond to mean $\pm$ SD for $n = 3$ devices.

Characterization of individual pairs of Electrofluidic Fiber Muscles

We characterized the quasi-static and dynamic response of electrofluidic muscle pairs under load and varying bias pressures in the system. In the experimental setup (Fig. 3A, fig. S5, and Supplementary Materials and Methods), McKibben actuators are fixed at the top to a vertical plate, with their lower ends attached to load holders connected to linear displacement sensors. Three-way valves connect the actuators to the fiber pump, and to experimental accessories: two syringe reservoirs (for filling the system and removing air bubbles) and two pressure sensors placed at the pump inlet and outlet. During testing, the valves are positioned in a way that the muscles are disconnected from the syringes and connected to the pressure sensors. Voltage is applied to the fiber pump using a compact 15-W high voltage power supply (0 – 10 kV) and monitored via a high voltage probe.

When voltage is applied, the fiber pump transfers fluid from the antagonist actuator (which extends) to the agonist one (which contracts), lifting the load. Figure 3B shows snapshots of the muscle pair at rest (left) and under steady actuation at 8 kV (right). After a 0.4 s transient, the agonist actuator contracts by 40 mm, lifting a 200 g load, while the antagonist extends by 23 mm. Movie S3 shows the operation of the experimental setup with live measurement.

We tested symmetric (equal loads on both sides of the muscle pair $m_C = m_E$) and asymmetric (contracting muscle loaded and extending one unloaded $m_E = 0$) loading conditions (Fig. 3C and D). Symmetric loading condition, due to its symmetry, was used for the physical model. Notably, even in the symmetric loading condition the net mechanical work generated by the contracting actuator during one actuation half-cycle is non-zero. Only a portion of the potential energy lost by the load m_E connected to the extending actuator is transferred through pressure energy to the contracting actuator and the lifted load (most of it gets dissipated).

a. Characterization varying bias pressure and voltage: physical model validation and experimental results

The physical model described in the previous section and Supplementary Text was used to estimate the steady-state contraction and response time of electrofluidic muscle pairs at different voltage and bias pressure values (Fig. 3E). To validate model results, we conducted experiments under the same conditions using the setup in Fig. 3. Both actuators were loaded with 200 g. Voltage ranged from 4 to 8 kV in 1 kV steps, and bias pressure from 0 to 125 kPa in 25 kPa increments. Voltage was applied from 0 to set value and response time was measured as 0 – 90 % of the agonist contraction.

Results show the model accurately predicts muscle pair response, generating a map that closely matches the experimental data. The model is fully physical: all parameters were based on physical quantities and no fitting multipliers were added. Discrepancies in peak strain and response time at low bias pressures are attributed to the model's simplicity, which excludes tube-braid friction in the McKibben actuators, viscoelastic expansion of the fiber pump and three-way valves under pressure, loads inertia (modeled as constant forces), and liquid inertia. These exclusions keep the model simple to understand and use. Despite its simplicity, model and experimental results show strong agreement across a wide operational range of electrofluidic muscles, confirming its validity.

Results show how the steady state contraction of electrofluidic muscles always increases by increasing the voltage applied to the fiber pump, while the response time decreases, for a

fixed bias pressure. Consistent with prior EHD and fiber pump literature, a higher voltage leads to both higher maximum pressure and flow rate (40, 46).

At a fixed voltage, both model and experimental data show that the steady state contraction stroke and strain are maximized for a bias pressure of 75 kPa, with this optimal bias pressure remaining constant across all voltage values. The existence of an optimum for bias pressure is a novel discovery and it is the result of shifting the application range of the fiber pump pressure towards a steeper region in the McKibben actuators' response curve (Fig. 2A, ii and Fig. 2C). Initially, increasing bias pressure leads to slightly increased response times, which then start decreasing. This trend is connected to the contraction stroke response, since for a given velocity of the muscle, the response time will be longer if the steady state contraction is higher. Response time variation with bias pressure, however, is of smaller entity than contraction variation.

A critical effect captured by both our model and experimental data is the existence of a cavitation region where electrofluidic muscles cannot be operated. The use of bias pressure allows to apply all the voltage range that the fiber pump can be operated at, thus enabling the full performance of electrofluidic muscles. At zero or low bias pressure values, voltages above 4 kV generate cavitation at the pump inlet, leading to the electrical breakdown of the pumps. Using the optimal bias pressure of 75 kPa or higher pressures, the system stably operates up to 8 kV, leading to a 37x increase in contraction stroke (37 mm vs 1 mm) and a reduction in response time over 50% (from 1 s to < 0.4 s). With 200 g load and applying an optimal bias pressure of 75 kPa, a pair of electrofluidic muscle at 8 kV achieves a contractile strain > 14% (stroke 40 mm) in 0.4 s (movie S3).

To further investigate cavitation and how bias pressure affects it, we measured fiber pump inlet pressure across different values of bias pressure and applied voltage. Figure 3F (iii) shows how the pressure at the pump inlet, for a given applied voltage, increases by increasing the bias pressure, as expected. For a given bias pressure, increasing the voltage leads to a lower inlet pressure. The reported inlet pressures correspond to the values reported in Fig. 3F (i) and (ii). It should be noted that measured inlet pressures exceed the filling bias pressure due to added mass (see Fig. 2E). When voltage is applied, the inlet pressure decreases and outlet pressure increases. Higher voltages lead to lower pressures at the inlet and higher ones at the outlet. The cavitation map in Fig. 3F (iv) shows inlet pressure curves for each bias pressure value. Cavitation occurs when voltage increases enough for the inlet pressure to drop below atmospheric (gauge $p < 0$), triggering electrical breakdown (movie S2). The cavitation threshold shifts higher with increasing bias pressure (Fig. 3F, i).

b. Characterization varying bias pressure and load at constant voltage

Figures 3G and 3H show steady state contraction and step response time of electrofluidic muscles under varying bias pressures and applied loads. As discussed earlier, increasing the load connected to the actuators increases the initial pressure in addition to bias pressure. Thus, the muscle can operate at low or even zero bias pressure when a high load is added to the actuators.

Figure 3G presents data for the muscle loaded symmetrically. Contraction and response time show similar trends as the ones in Fig. 3F, with contraction peaking at an optimum bias pressure of 75 kPa, independent from the applied load. This counterintuitive effect results from a combination of opposite effects: (1) greater loads increase initial pressures for given bias pressure, (2) increasing loads shifts the high response region of the McKibben

contraction-pressure curve toward higher pressures (Fig. 2A, ii). These two effects balance out, yielding a constant optimal bias pressure of 75 kPa across tested loads (100 – 500 g). For a given bias pressure, increasing the load decreases both contraction and response time. Figure 3H reports the same tests but with the muscle loaded asymmetrically. Loads from 100 to 500 g are applied on the contracting actuator only, while the extending one is left unloaded. The global trends of contraction and response time dependencies with bias pressure are also shown for asymmetric loading. Contraction/strain and response time do not differ significantly from the ones for the symmetric loading case. Response times are on the order of 2x longer, since the mass on the extending muscle in the symmetric case was partially contributing to lifting the load on the contracting muscle through pressure energy.

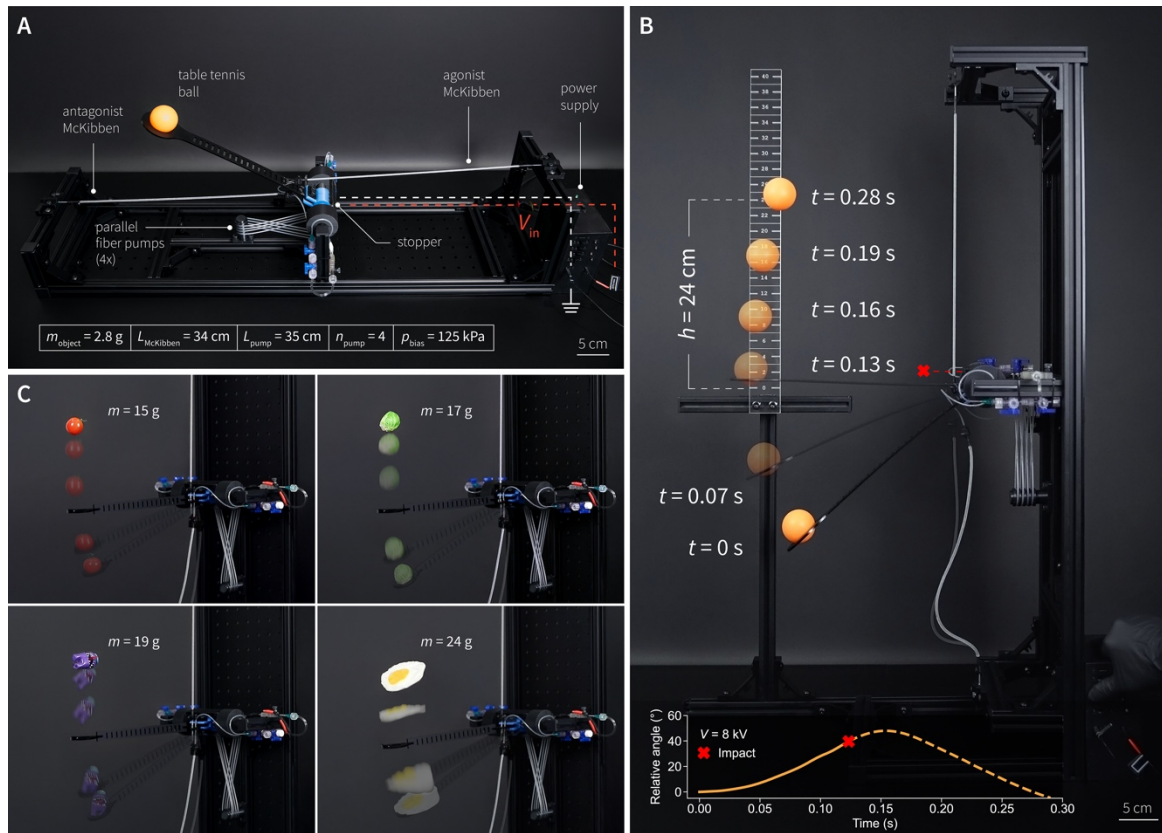


Fig. 4. Four fiber pumps connected in parallel demonstrating rapid actuation. (A) A lever arm / catapult setup was connected to an antagonistic pair of McKibben actuators and four fiber pumps arranged fluidically in parallel configuration to increase flow rate and hence shorten response time. (B) When 8 kV is applied to the muscle, the agonist actuator contracts rapidly within 0.13 s, driving the lever arm into the horizontal stopper and launching the table tennis ball 24 cm upward in < 0.3 s. Time measurements start from the onset of muscle contraction ($t = 0$ s). (C) Everyday objects of various masses (< 25 g) are rapidly launched into the air.

Demonstrating rapid actuation: four fiber pumps connected in parallel driving two McKibben actuators

Electrofluidic fiber muscles, operating under an optimal bias pressure, exhibit performance that enables dexterous robotic applications. Programming performance output is simply done by selecting the number of fiber pumps and McKibben actuators arranged in each muscle. To achieve fast response, we connected four fiber pumps fluidically in parallel to an antagonistic pair of McKibben actuators. A high ratio of pumps in parallel over actuators leads to high flow transfer into the actuators in a short time. The resulting muscle responds is fast enough to throw various objects in the air when connected to a lever arm (Fig. 4A, fig. S9, movie S5).

Arranging multiple pumps in parallel effectively increases the flow rate and instantaneous power density in the system. When we apply a voltage step (8-kV), the agonist muscle contracts fully in 0.13 s, achieving a contraction rate of 180 mm/s (Fig. 1E, iv and fig. S10). We arranged a stopper to block the lever arm when in a horizontal position. The arm effectively throws a table tennis ball 24 cm into the air (Fig. 4B). Movie S6 and Fig. 4C show the lever arm response, throwing a series of objects: a table tennis ball, a cherry tomato, a brussels sprout, and flipping a fried egg prop and ‘Among Us’ toy figurines.

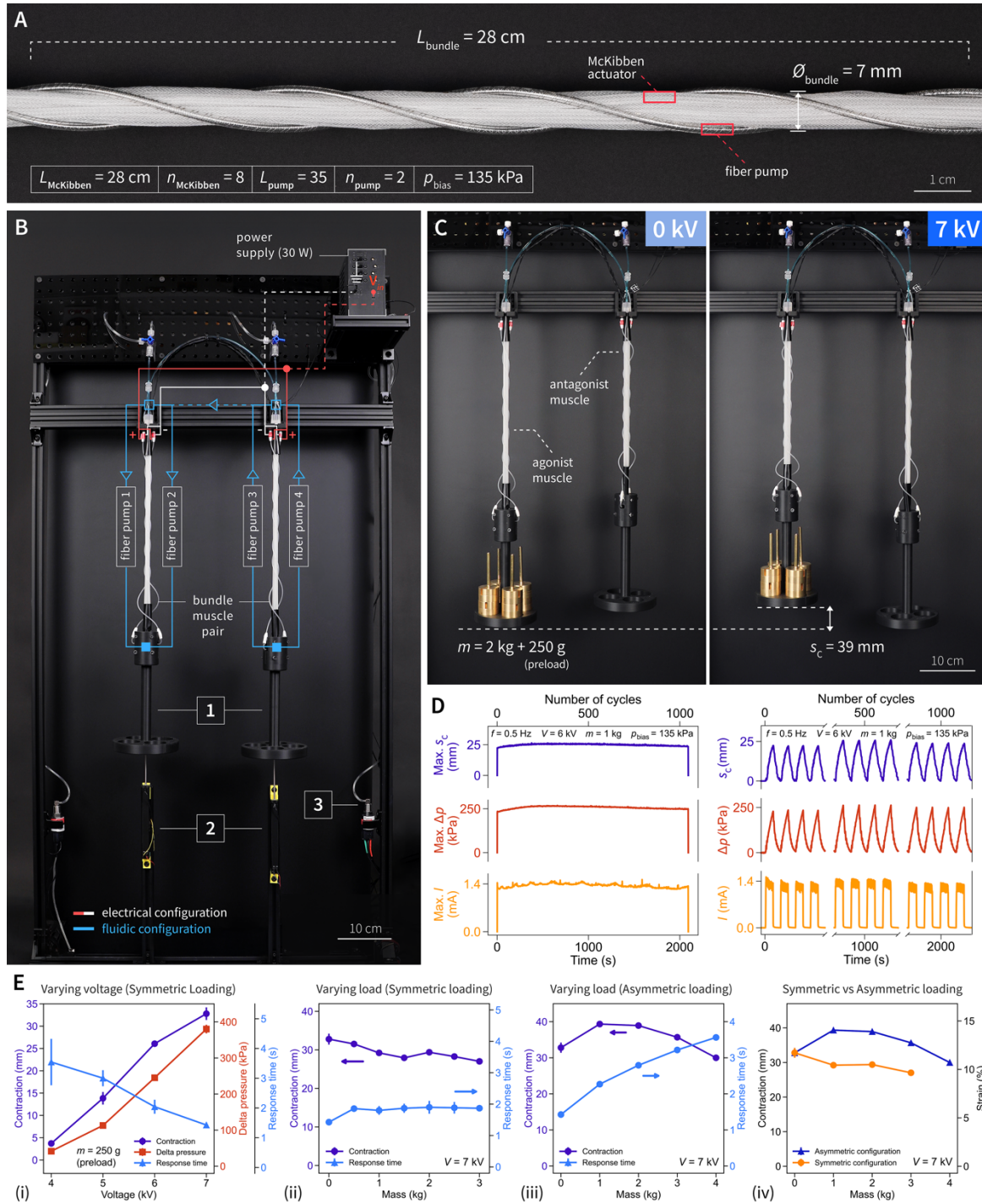


Fig. 5. High force electrofluidic muscle bundles lifting > 4 kg. (A) Close-up of one muscle bundle consisting of 2 fiber pumps ($L = 350 \text{ mm}$), wound around 8 McKibben actuators ($L = 280 \text{ mm}$). (B) Electrical and fluidic configurations of muscle bundle pair arranged to scale up force generation in a compact form factor. The characterization setup includes (1) two load holders, (2) two displacement sensors, and (3) two gauge pressure sensors monitoring inlet and outlet pressures. (C) Muscle bundle under a 2-kg (+250 g preload) asymmetric load, contracting by 13.6 %. (D) Cyclic stability test for 1000 cycles at 0.5 Hz, lifting 1 kg (+250 g preload) on each bundle at 6 kV, and 135 kPa of bias pressure. Contraction (s_c), delta pressure (Δp) and current (I) over time. Maximum currents evaluated after applying a low-pass filter (cutoff frequency: 10 Hz) on the raw data. (E) Muscle performance under symmetric loading with voltage varying between 4 – 7 kV (preload) (i); contraction and response time vs. additional mass (ii); under asymmetric loading, contraction and response time vs. additional mass (iii); comparison of contraction and strain between symmetric and asymmetric loading conditions (iv). Values with error bars correspond to mean $\pm$ SD for 10 cycles.

Electrofluidic Muscle Bundles for scaling up forces: two parallel-two series pumps driving antagonistic 8-McKibben bundle actuator pair

We bundled fluidic muscles to demonstrate scalable force generation by increasing the number of McKibben actuators in parallel, following design principles of Suzumori et al. (47, 48), analogous to mammalian muscle architecture. Each muscle pair consists of two identical antagonistic bundles. Each bundle is made of eight thin McKibben actuators ($ID = 1.25$ mm, $OD = 1.75$ mm) arranged in a circular geometry, with two fiber pumps wound around them ($OD = 7$ mm, $L = 280$ mm) (Fig. 5A and movie S7). McKibben actuators on each side are arranged in a fluidically parallel configuration, sharing the same fluidic input. Two fiber pumps on each side are also configured in parallel, effectively increasing the flow rate. The two bundles are then fluidically connected in series, forming a closed fluidic circuit (Fig. 5B, fig. S9 and Supplementary Materials and Methods). In this series configuration, the pressure differential generated on each muscle is the sum of the one generated by each pair of pumps in parallel (roughly double the pressure of a single pump). The resulting device is a light, flexible and compact robotic muscle weighing only 22 g (two 8-McKibben bundles + 4 fiber pumps filled with liquid), capable of lifting 200 times its own weight (4 kg + 250 g preload), and achieving 30 mm contraction (11 % strain).

Muscle bundles were tested under two loading configurations—symmetric and asymmetric—at a bias pressure of 135 kPa. This bias pressure was selected based on two key considerations: 1) the model-predicted optimal bias pressure for this demonstration, and 2) a conservative cavitation threshold for safe operation, determined by monitoring inlet pressure during actuation tests (fig. S11).

Figure 5C shows the pair in asymmetric loading configuration (only the contracting/agonist muscle is loaded). When voltage is applied to the four pumps (all electrically in parallel), the agonist bundle contracts, lifting the connected load of 2.25 kg (39 mm stroke), while the antagonist extends.

Long-term cyclic operation (Fig. 5D) demonstrated stable performance over 1000 cycles at 0.5 Hz under 1 kg per bundle with 6 kV applied voltage.

Figure 5E shows the electrofluidic muscle bundle characterization. The bundle was first tested in the symmetric configuration by increasing the applied voltage while measuring the response with only the preload. A contraction of 32 mm was achieved within 1.2 seconds, with a pump-generated pressure differential of 400 kPa at 7 kV (Fig. 5E, i). Next, increasingly higher loads are added symmetrically to both muscles, with a maximum load of 3 kg per muscle. Under these conditions, a contraction of 27 mm was measured in 1.8 seconds at 7 kV (Fig. 5E, ii). At the same voltage, the system was also tested in asymmetric loading configuration, supporting a maximum load of 4 kg (Fig. 5E, iii and movie S8). This resulted in up to a 30% increase in contraction compared to the symmetric case, with consequently longer response time. Figure 5E (iv) shows the comparison between symmetric and asymmetric loading.

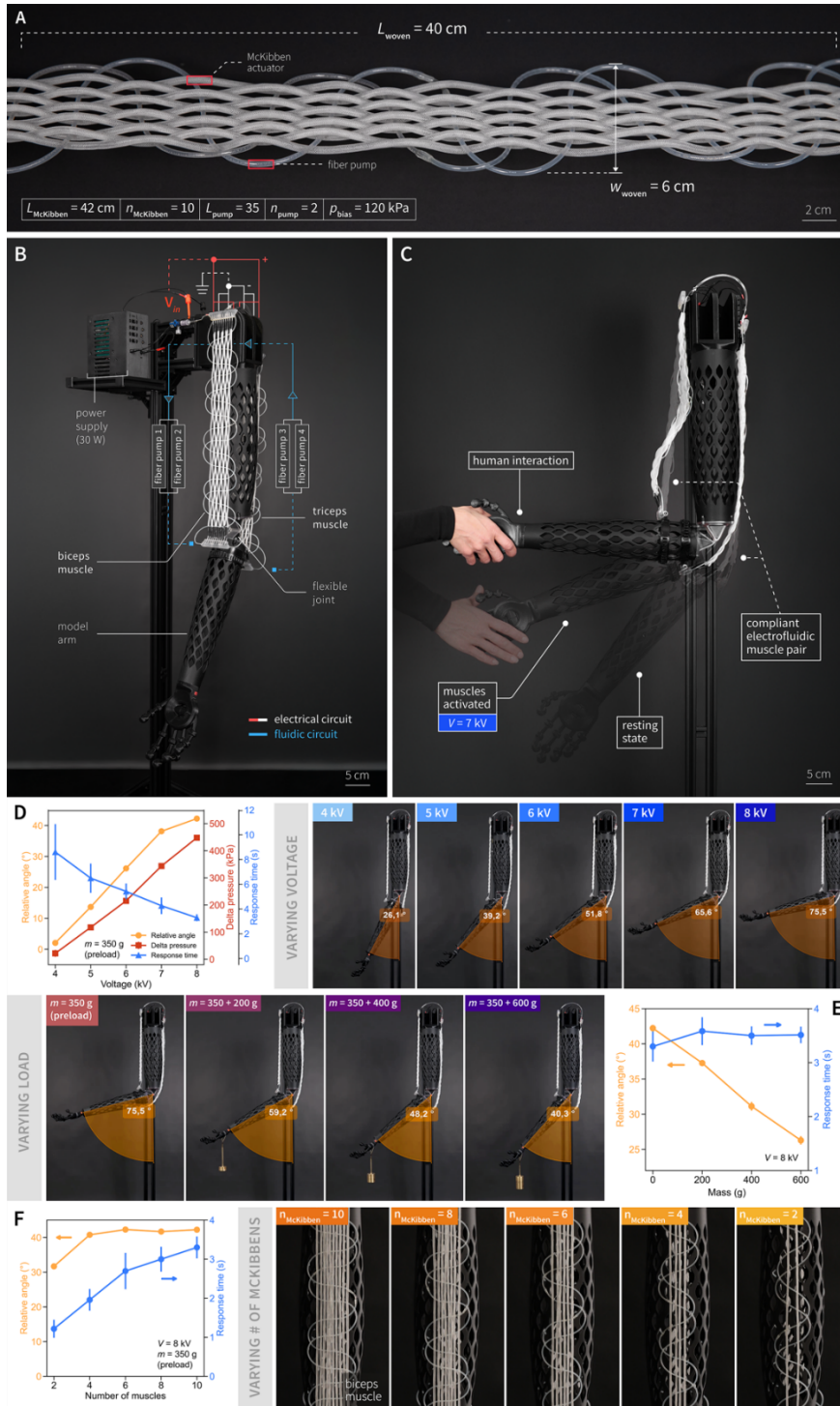


Fig. 6. Back drivable electrofluidic woven muscles for compliant wearable robotic applications. (A) Close-up of one woven muscle consisting of 2 fiber pumps ($L = 350 \text{ mm}$) woven with 10 McKibben actuators ($L = 420 \text{ mm}$). After weaving the actuators and pumps together, woven muscle length is 40 cm and width is 6 cm. (B) Electrical and fluidic configurations of a woven biceps and triceps muscle pair driving a human-scale forearm model. The forearm (preload) weighs 350 g. (C) Fluidic muscles are compliant and back-drivable, enabling smooth human-robot interaction. (D) Arm bending is proportional to applied voltage at fixed load. (E) The muscle lifts up to 600 g additional mass with a relative bending angle of 26 degrees at 8 kV. (F) Demonstrating scalability and modularity: reducing the number of McKibben actuators in the biceps from 10 to 4 shortens the response time by 42 %, while the relative bending angle remains unchanged ($> 40^\circ$), at fixed load. For the whole figure: values with error bars correspond to mean $\pm$ SD for 10 cycles, applied bias pressure is 120 kPa, relative angle refers to the difference between the final angle of the arm in active state and initial angle at rest.

Electrofluidic Woven Muscle pair with a large range of motion for wearable robots: two parallel-two series pumps driving antagonistic 10-McKibben sheet actuator pair

A key advantage of 1-mm fiber muscles is that we can intertwine them into textiles. To demonstrate this, we created flat textile muscle pairs weaving McKibben actuators and fiber pumps together (Fig. 6A). The resulting woven muscles are flexible and ready for integration to textile exosuits that offer muscular support. Their design is highly modular: the number of McKibben actuators in parallel sets the load capacity of the muscle, while the ratio between McKibben actuators and number of fiber pumps in parallel sets the response time. The woven muscle we report consists of ten parallel McKibben actuators ($ID = 0.7$ mm, $OD = 1$ mm) forming the warp, and two fluidically parallel fiber pumps interwoven as the weft (Fig. 6A and fig. S9). A 40 cm bare TPU extension tubing was added to each pump to span the entire length of the muscle. The resulting woven muscle is 40 cm long and 6 cm wide (movie S9 and Supplementary Materials and Methods).

We demonstrated the electrofluidic woven muscles as biceps and triceps pair of a 3D printed human-scale robotic arm (Fig. 6B). Woven muscles can be easily bent, folded, twisted and stretched (movie S9). Operating under bias pressure, they remain compliant and intrinsically back stretchable in all configurations. We demonstrate this important feature in a human-robot handshake (Fig. 6C and movie S10). Muscles drive the robot arm to a handshake position. The human can naturally shake it since both muscles easily deform: biceps go slack, and triceps stretch. As demonstrated by years of research in legged robots, back drivable actuators are essential to create agile robots (49). Common EM motors for robots are not back drivable, due to the high gear ratio required to achieve high torque with small sizes. While large-radius EM motors and additional mechanisms have been explored for back drivability, they are too bulky for wearables (50). McKibben actuators alone require extra components to allow back-stretchability at rest (44). Thanks to bias pressure, our electrofluidic muscles offer intrinsic compliance and back-stretchability in all their operating conditions, paving the way to dynamic human-robot interaction and untethered soft wearable robots (movie S11).

Figures 6D and 6E show woven muscles characterization on the robotic arm. In these tests, we selected a bias pressure of 120 kPa, based on three key considerations: (1) the model-predicted optimal bias pressure for this configuration (105 kPa), (2) the cavitation threshold for safe operation, determined by monitoring inlet pressure during preliminary actuation tests, and (3) to prevent reaching McKibbens' burst pressure (550 kPa) (fig. S11). As expected, the bias pressure in this system is higher than the value predicted for one pump (75 kPa, see Fig. 2-3), since here we have two pumps in series (also two pumps in parallel but adding pumps in parallel does not influence pressure) (Fig. 6B).

Figure 6D shows the bending response of the robotic arm driven by the woven muscle pair without additional mass (forearm $m = 350$ g). By increasing the voltage from 4 to 8 kV, the arm bends at proportionally larger angles, reaching $> 40^\circ$. The maximum relative angle of 42° was reached in 3.2 seconds at 8-kV, driven by a pump-generated pressure of 460 kPa (movie S10). We then tested the device at a fixed voltage of 8 kV while hanging masses up to 600 g on the robotic hand (Fig. 6E). Under a 600 g mass, the biceps muscle generated 30 N force (fig. S10).

This woven setup exhibits a significantly slower response than the individual muscle pairs and the lever arm due to the lower ratio between fiber pumps in parallel and McKibben actuators. We demonstrate the effect of modular tunability by adjusting this ratio on the

same system. Figure 6F and movie S10 show the response of the system by varying only the number of McKibben actuators in the biceps. Moving from the standard configuration with 10 actuators to one with 2 actuators (keeping the same number of pumps), the response time reduces from > 3 s to ~ 1 s, with only a small reduction in the bending angle. While bending angle would eventually decrease at higher loads with fewer actuators, this result highlights that tuning the actuator-to-pump ratio enables control over response time to meet target specifications.

DISCUSSION

In this article, we report on self-contained electrofluidic muscles, a class of soft actuators that is electrically driven yet uses a liquid as a coupling medium. The liquid is pressurized by 1-2 mm sized fiber pumps and transferred into thin McKibben actuators, producing contraction. We studied and designed the operation of these muscles in a closed system: using an antagonistic configuration and an optimal bias pressure it is possible to remove an external liquid reservoir and to dramatically boost performance, increasing coupling efficiency and preventing cavitation. We report the use of these muscles in untethered demonstrators, whose shape, size, and performance have been designed by choosing the number of actuators, pumps, and their arrangement. Our demonstrators generate large forces (> 4 kg) and fast response (< 0.3 s), with a measured average power densities exceeding 50 W/kg.

Some limitations for practical deployment exist. The main one is the low energy efficiency of fiber pumps, which is due to a still limited understanding of the EHD mechanism. Also, the liquid that is currently used is a FluoroHydroCarbon. It is non-toxic and commercial, yet it is provided by only one manufacturer so its availability might hinder large scale deployment of these artificial muscles. Finally, McKibben actuators show non-negligible hysteresis in their cyclic operation which may be compensated for by bidirectional operation of fiber pumps in future work.

A deeper understanding of EHD and the use of a wider range of commonly available liquids will pave the way for the use of electrofluidic fiber muscles to build a new generation of highly dexterous robots and wearables.

MATERIALS AND METHODS

A short summary of design, materials, fabrication, characterization setup, experimental protocols and statistical analysis are provided in this section. Further details for each sub-section are reported in Supplementary Materials and Methods.

McKibben actuators materials and fabrication

Thin McKibben actuators with varying diameters and lengths were fabricated following previously established protocols (fig. S4, A and B).

McKibben actuators are composed of an internal elastomer tube and an outer braid that guides its deformation. We fabricated our actuators using Thermoplastic Urethane (TPU) tubing, with hardness shore 40A and a wall thickness of 0.2 mm. TPU was chosen over the more widely used silicone rubber because of its compatibility with the fluorinated dielectric oil used in fiber pumps. The outer braid is composed of 100 μ m thick nylon monofilaments ($n = 24$) with an initial braid angle of 17° (Fig. 2A, i).

Fiber pumps materials and fabrication

The fiber pumps used in this study are based on the design and fabrication method presented by Smith et al. (40). The pumps we fabricated have a shorter electrode distance (0.5 mm) and a slightly larger helix angle (63 degrees).

Fiber pumps were fabricated via filament winding using platinum electrodes and TPU monofilaments, forming a double-helix structure with ~ 0.5 mm electrode spacing (fig. S3). All pumps had an inner diameter of 1.1 mm, outer diameter of 1.6 mm, and length of 350 mm.

We used as electrode material platinum wires (75 μm diameter) and as tube polymer TPU 95A. Figure 2B shows (i) a close-up image of one fiber pump, with the spiral shaped electrodes visible, and (ii, iii) characterization results of three identical fiber pumps with $ID = 1.1$ mm, $OD = 1.6$ mm, $L = 350$ mm.

For all the experiments, 3M Novec 7100 (methoxy-nonafluorobutane $\text{C}_4\text{F}_9\text{OCH}_3$) is used as the dielectric fluid.

Design of closed-circuit Electrofluidic Fiber Muscles

Electrofluidic fiber muscles are composed of one or more fiber pumps (Fig. 1A, i), connected at both ends to one or more thin McKibben actuators (Fig. 1A, ii and iii) that are sealed at their distal ends. We designed the muscles as a closed system without the need for an external reservoir, enabling mobile, untethered operation on robots and wearables (Movie 1 and movie S11). Once the system is filled with dielectric liquid, the total liquid volume in the circuit (pump and actuators) does not change during operation. Each actuator acts as a reservoir for its antagonist, making electrofluidic muscles compact and portable.

The system is activated by applying a voltage difference to the two electrical terminals of the fiber pump (Fig. 1C, i), that, drawing energy from the power supply, moves liquid from one actuator, causing it to extend, and transfers it to the other, which contracts (Fig. 1C, ii, Movie 1 and movie S1). When initially filling the muscle with dielectric liquid, we add an extra liquid volume to pre-pressurize the system at a bias pressure that we can regulate. This design offers several advantages: a) each McKibben actuator acts as a liquid reservoir for the other (no additional reservoir required), b) as in mammalian muscles, an antagonistic configuration (42) matches well the actuation of a rotary joint, with one actuator that pulls one side of the link while the other extends following the rotation, c) the pressure energy stored in a pressurized actuator is reused when reversing motion and actuating the opposite one, shortening actuation time, d) the bias pressure of the system, i.e., pressure in the closed circuit when the fiber pump is not activated, can be regulated, dramatically impacting performance.

Natural muscles are organized as structured bundles or sheets, rather than individual fibers. We designed diverse ensembles of electrofluidic fiber muscles by varying the number and arrangement of fiber pumps and McKibben actuators, tuning system performance in terms of force (up to 42 N), velocity (up to 180 mm/s), and range of motion (Fig. 1E, F).

Experimental setup and protocols

Characterization of McKibben actuators, fiber pumps and individual muscle pairs (one fiber pump connected to two McKibben actuators)

The experimental setup shown in Fig. 3A involves two McKibben actuators in an antagonistic configuration, driven by one fiber pump connected between them. The muscle was actuated using a compact high-voltage DC power supply (0–10 kV, 1.5 mA), and pressure differential across the pump was measured via gauge sensors at the inlet and outlet. Displacement of each actuator under both symmetric and asymmetric loading conditions (100–500 g) were recorded using linear potentiometers. For fiber pump characterization, flow rate was measured by integrating a flow meter in parallel with the pump (fig. S5A). All data—including pressure, displacement, flow rate, voltage, and current—were recorded using two synchronized DAQ devices at 500 Hz. The circuit was filled with dielectric fluid and sealed before each test.

a. Characterization of McKibben actuators

We evaluated the displacement response of vertically mounted McKibben actuators ($n = 3$) under varying bias pressures and loads on a simplified version of the experimental setup with a gauge pressure and linear displacement sensor. A syringe pump injected dielectric fluid at a controlled rate, and actuator deformation was recorded using the same DAQ device.

These actuators contract when pressurized. We characterized their contraction as a function of pressure and volume for actuators with 1.5 mm outer diameter (*OD*), when lifting different loads. The resulting curves (Fig. 2A, ii and iii) show the characteristic pressure-contraction and volume-contraction responses of these actuators. Here, both pressure and volume are considered relative to the baseline values when the actuators are filled with liquid and undeformed. The pressure-contraction response curve shows initially a flat response, especially under high loads, while the response steepens at higher pressure values. This nonlinear behavior, typical of McKibben actuators (43), results in small contractions until the steep response region is reached. In the final part of the pressure range the response tends to flatten again as maximum contraction is reached.

An analysis of these nonlinear response curves is essential for an effective coupling between McKibben actuators and fiber pumps. To date, fluidic actuators including McKibbens have been mostly operated with large external pumps. These pumps generally include a large tank and proportional valve, which provide a constant pressure output. The pressure values that these large pumps generate can easily span the entire pressure range of McKibben actuators operation. When using integrated and miniaturized fiber pumps, we need to account for the fact that the pump pressure is finite and limited. Therefore, it becomes essential to make the best use of the pressure that the pump can generate, operating the muscles in their steepest response region. The region of operation can be simply changed by selecting the bias pressure that the muscles are operated at, as shown in Fig. 2C. While increasing the pressure that fiber pumps can generate can be challenging (requires novel insights in EHD or fabrication of longer pumps that take up more space), selecting a given bias pressure is simply done by filling the muscles with more dielectric liquid before operation. Operating in the steep region of the pressure-contraction response curve means a higher contraction can be achieved for a given pressure output of fiber pumps.

The volume-contraction response curve follows a similar trend (steeper responses at higher volumes) even though with less pronounced changes compared to the pressure-contraction curves. Therefore, a similar analysis of coupled-system design can be done for the limited flow rate of integrated and miniaturized fiber pumps compared to large external pumps. Using the steepest regions of the volume-contraction response curve leads to faster response for a given flow rate of fiber pumps.

b. Pressure - flow characterization of fiber pumps

We characterized fiber pumps ($n = 3$) under varying pressure conditions (open/closed circuit, with and without bias pressure), using fresh dielectric liquid for each test. Pump flow rate and pressure were measured across voltage steps from 1 to 8 kV.

The devices were tested under three configurations of fluidic circuit: (1) constant atmospheric pressure (open circuit, the inlet of the pump is connected to a liquid reservoir open to the environment), (2) closed circuit (no fluid reservoir) with no bias pressure applied ($p_{\text{bias}} = 0$ kPa), and closed circuit with extra liquid added until reaching a bias pressure $p_{\text{bias}} = 75$ kPa.

The results show pressure and flow rate increasing more than linearly with voltage. Performance at the same voltage is independent from the fluidic circuit configuration. However, the configuration with the closed circuit and no bias pressure ($p_{\text{bias}} = 0$ kPa), could be tested only until 4 kV due to electrical breakdown triggered by cavitation of the dielectric liquid. Both open circuit and closed circuit with $p_{\text{bias}} = 75$ kPa could be tested until 7 kV and showed same performance. The configuration with $p_{\text{bias}} = 75$ kPa could be safely tested until 8 kV, showing good stability and producing a maximum delta pressure of 320 kPa (Fig. 2B, ii) and a maximum flow rate of 95 mL/min (Fig. 2B, iii).

c. Characterization of individual muscle pairs (one fiber pump connected to two McKibben actuators)

Performance of muscle pairs were characterized under varying voltage, bias pressure, and load conditions. Muscle pair mass measurements are provided in table S2. Each test was performed on $n = 3$ devices, with five actuation cycles at 0.05 Hz per device. Prior to cyclic actuation, the applied voltage was held constant for 30 seconds to allow stabilization. Data points represent 10-second averages per cycle, further averaged across devices.

d. Dynamic characterization of individual muscle pairs

The dynamic tests consisted of varying frequency from 0.25 Hz to 10 Hz at a fixed condition of 75 kPa bias pressure at 7 kV. The power supply output was adjusted and allowed to stabilize for 30 seconds before initiating the cyclic tests, after which each frequency was applied for 20-second-long cycles per device, across three identical devices.

e. Characterization of robotic applications

For all demonstrations in Figs. 5 to 6, we applied the driving signals with a compact custom 30-W power supply with an embedded DC converter (Advanced Energy, 10A24-P30-I5) shown in fig. S8. For the lever arm demo in Fig. 4, we analyzed jump heights and corresponding time stamps by recording motion with a high-speed camera (Sony Alpha 7 III) at a frame rate of 120 fps. For the bundle demo in Fig. 5, we measured displacements using linear potentiometers and response time was captured using a dual DAQ (Digilent Discovery MCC-234) setup. For the woven demo in Fig. 6, response times and bending

angles were determined by analyzing arm motion recorded with a Canon EOS 5D Mark IV camera using Kinovea (2023.1.2) analysis software.

f. Statistical analysis

Muscle pair tests were performed on $n = 3$ devices over five cycles per condition, and robotic applications used a single device with 10 cycles per data point (mean $\pm$ SD). Data were processed using Python, with signals down sampled to 50 Hz. Duty cycles were segmented using electrical signals.

Supplementary Materials

Supplementary Materials and Methods

Supplementary Text

Figs. S1 to S11

Tables S1 and S2

Legends for Movie 1 and movies S1 to S11

References (51–60)

Movies S1 to S11

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Supplementary Materials for

Electrofluidic Fiber Muscles

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The Supplementary Materials include:

Supplementary Materials and Methods

Supplementary Text

Figs. S1 to S11

Tables S1 and S2

Legends for Movie 1 and movies S1 to S11

References (51–60)

Other Supplementary Material for this manuscript includes the following:

Movies S1 to S11

SUPPLEMENTARY MATERIALS AND METHODS

Fabrication of fiber pumps

Fiber pumps were fabricated using a filament winding approach adapted from Smith *et al.* (40). Six thermoplastic polyurethane (TPU) monofilaments (Recreus Filaflex 95A) and two platinum (Pt) electrode wires were wound together around a stainless-steel mandrel to form the pump body and embedded double-helix electrodes.

Prior to winding, TPU filament was extruded through a 0.4 mm nozzle at 235 °C at high speed, yielding monofilaments with a final diameter of ~300 μm . The Pt electrode wire, 75 μm in diameter, was selected for its corrosion resistance and long-term electrochemical stability similar to other noble metals such as gold (51, 52). Following extrusion and cutting, 60 g and 30 g weights were attached to the ends of the TPU and electrode filaments, respectively, to maintain consistent tension during winding. The mandrel and Pt wires are cleaned with isopropyl alcohol (IPA). The mandrel is then coated with a thin layer of mold release agent (Mann Release 400) to enable easy pump removal after the fusion process.

The winding machine (fig. S3A) consists of a rotary stage coupled to a linear actuator. A 3D-printed guide mounted on the rotary stage directs the winding of TPU and electrode filaments along a repeatable path onto the stainless-steel mandrel. As the filaments are tightly wound, the electrode wire settles beneath the TPU monofilaments, resting directly on the mandrel surface. Continuous winding maintains tension in the TPU, which secures the helical structure and prevents displacement of the embedded electrodes during fabrication (fig. S3B). The winding process is monitored using a digital microscope (Dino-Lite AM3111) positioned a few millimeters above the winding guide to observe filament and wire placement onto the mandrel in real time. Angular and linear velocities are programmatically synchronized to achieve a consistent monitored winding angle of ~63° (fig. S3C). At the target pump length, winding was paused to withdraw the electrode wire through the guide, creating a minimum of 2 cm electrode-free section at one end. At the opposite end, the electrode wire was manually unwound over ~2 cm to allow for TPU-only pump ends. Both ends were secured with cable ties to hold the tension prior to and during thermal fusion. After securing the ends of the pump with cable ties, the pump was removed from the winding machine and prepared for the Joule heating based fusion process. The fusion step is critical in fiber pump fabrication, as it forms a sealed tube that electrically insulates the external electrode surface while maximizing internal electrode wire exposure and preserving mechanical robustness. Fusion was achieved via Joule heating by passing current through the stainless-steel mandrel used during winding. A potential difference applied across the mandrel ends (DC power supply) induced resistive heating, raising the mandrel temperature to ~180 °C. Prior to fusion, a small amount of UV-curable adhesive was applied at the four wire exits on both ends of the pump to secure the electrode wires in place during heating. Before testing, high-voltage (HV) wires—cut to the desired length—were brought into contact with the exposed electrode segments using silver conductive epoxy (MG Chemicals 8331D-14G). All connections were subsequently insulated using UV-curable adhesive (Norland NOA61) and heat-shrink tubing to prevent electrical arcing.

All fiber pumps manufactured for this study featured an inner diameter (*ID*) of 1.1 mm, outer diameter (*OD*) of 1.6 mm (fig. S3D), and electrode spacing of 500 μm . The simple winding technique allows for the fabrication of fiber pumps with varying characteristics. Specifically, the inner diameter of the pump can be modified by selecting mandrels of different diameters, and the electrode spacing can be adjusted by varying the number of

TPU filaments between the wires all of which have effect on the EHD performance. Further, our linear stage allows for manufacturing pumps up to a maximum length of 70 cm (fig. S3E).

Fabrication of thin McKibben actuators

Thin McKibben actuators were fabricated using Shore 40A thermoplastic polyurethane (TPU) tubing (Chronoprene 40A), selected for its chemical compatibility with 3M Novec 7100, the dielectric fluid used in the experiments. The fabrication process is modular and can be readily adapted to McKibben actuators with varying geometries and dimensions. First, the TPU tubing was cut to the desired length. Stainless steel blunt needles (Component Supply, HTX-C075-18R) were then inserted into both ends of the tubing, and a UV-curable adhesive (Norland NOA61) was applied and cured at the tubing-needle interfaces to achieve a fluid-tight seal. The sealed tubing was subsequently inserted into a braided sleeve (Techflex, FlexoThin, $ID = 1.6$ mm). The nylon braid was secured to the tubing near each end using additional UV-curable adhesive, and any excess braid was trimmed away. Heat-shrink tubing was applied over the bonded regions to protect the adhesive joints while leaving a defined active length exposed for actuation. To enable fluidic interfacing, segments of Shore 95A connector tubing ($ID = 1$ mm, $OD = 1.5$ mm) were press-fitted onto the inlet and outlet needles, allowing connection to commercially available fittings such as luer-lock needles (fig. S4A).

Following fabrication, each actuator underwent a liquid leakage test prior to integration with fiber pumps. During operation, one luer outlet was sealed with a luer cap, while the opposing end was connected to a pressure source (e.g., fiber pump) to drive fluid into the actuator. In this study, three different McKibben geometries were used (fig. S4B). The largest actuator, with an inner diameter (ID) of 1.25 mm and outer diameter (OD) of 1.75 mm, was used in the muscle bundles demonstration (fig. S4B, i). The second actuator, with an ID of 1.1 mm and OD of 1.5 mm, was used for single McKibben and muscle pair characterization as well as the lever arm demonstration (fig. S4B, ii). The thinnest actuator, with an ID of 0.7 mm and OD of 1.0 mm, was used in the woven muscles demonstration (fig. S4B, iii).

Performance characterization of McKibben actuators, fiber pumps and individual muscle pairs (2 McKibben actuators + 1 fiber pump)

Experimental setup

The experimental setup shown in Fig. 3A involves two fiber muscles in an antagonistic configuration, driven by one fiber pump connected in between them. The pump is activated by a custom compact DC 15 W high-voltage power supply (0 – 10 kV, 1.5 mA) shown in fig. S8. Two gauge pressure sensors are connected respectively to the inlet and the outlet of the pump, so the pressure difference generated by the pump is measured as the difference between the gauge values of the outlet and inlet pressure. The muscles are attached to a vertical plate and their bottom ends are connected to 3D printed holders for loads of different mass (100 – 500 g), which in turn are connected to two vertical linear potentiometers (P3 America LMC13 Series) used as displacement sensors to measure the stroke of each muscle. When a DC voltage is applied to the fiber pump, the agonist actuator contracts and lifts the attached load, while the antagonist actuator relaxes and extends. For the fiber pumps'

characterization, a flow meter (Sensirion SLF3S-4000B) is added in parallel to the pump (fig. S5A).

All data (inlet and outlet pressure, contraction and extension strokes, flow rate, applied voltage, and current) are recorded using two Digilent Discovery MCC-234 DAQ devices in differential mode reading at a sampling rate of 500/s. The two DAQ devices are synchronized via a shared clock.

Before each test, the system was filled with dielectric fluid using two polypropylene syringes, one on each end of the fluidic circuit. Two syringes are used to clear the system of air bubbles. Extra liquid is then added to reach the desired bias pressure in the system and 3-way fluidic valves are closed to stabilize the bias pressure. Before starting the test, loads of different masses are applied. Across the article, we refer to applied mass as the total external mass added to a single McKibben actuator. Applied mass equals the sum of the preload (load holder) and additional mass (extra loads added on the load holder). During the test, a dual DAQ system manages voltage application and data collection.

Data measurements and performance evaluation

Contraction (s_C) refers to the contractile stroke of the agonist McKibben, achieved from the pre-pressurized state (bias pressure baseline) to the actuated state.

Elongation (s_E) refers to the extensile stroke of the antagonist McKibben, achieved from the pre-pressurized state (bias pressure baseline) to the actuated state.

We measure the strain (%) as:

$$\text{strain} = \frac{s}{l_o} \times 100$$

where s is the stroke (contractile or extensile) and l_o is the length of the McKibben actuator in the undeformed state (no load applied, $p = 0$ kPa).

We evaluate the response time (t_{90}) as the time required by the agonist McKibben to reach 90% of the maximum contraction (s_C), starting from the pre-pressurized state (bias pressure baseline) at time 0, when the voltage is applied.

We measure the velocity (mm s^{-1}) as:

$$\text{velocity} = \frac{s_C}{t_{90}}$$

where s_C is the contractile stroke and t_{90} is the response time.

We evaluate the strain rate ($\% \text{ s}^{-1}$) as:

$$\text{strain rate} = \frac{\text{strain}}{t_{90}}$$

where t_{90} is the response time.

Outlet pressure (p_{out}) refers to the value of pressure achieved in the agonist McKibben in the actuated state.

Inlet pressure (p_{in}) refers to the value of pressure achieved in the antagonist McKibben in the actuated state.

Delta pressure (Δp) refers to the pressure differential generated by the fiber pump in response to applied voltage. We evaluate Δp (kPa) as:

$$\Delta p = p_{\text{out}} - p_{\text{in}}$$

Energy density (E_d) is computed as the ratio between the final end-effector mechanical work and system mass, dependent on the applied bias pressure. We evaluate E_d (J kg⁻¹) of the system as:

$$E_d = \frac{m_C g s_C}{m_s}$$

where m_C is the mass of the load lifted by the agonist McKibben, g is the gravitational acceleration, s_C is the contraction stroke of the agonist muscle, m_s is the mass of the system as filled with dielectric fluid. The mass of the system includes: (1) two McKibben actuators and one fiber pump filled with liquid (the liquid volume and total muscle mass change according to applied bias pressure as shown in table S2); (2) two stainless steel blunt needles, for connection between pump and McKibbens; (3) two 10 cm-long HV leads.

Power density (P_d) is computed as the ratio between average mechanical power output and system mass. We evaluate P_d (W kg⁻¹) of the system as:

$$P_d = \frac{m_C g s_C}{m_s t_{90}}$$

where m_C is the mass of the load lifted by the agonist McKibben, g is the gravitational acceleration, s_C is the contraction stroke of the agonist muscle, m_s is the mass of the system as filled with dielectric fluid (table S2) and t_{90} is the response time 0 – 90%.

We evaluate the instantaneous power (P_{inst}) as:

$$P_{\text{inst}}(t) = m_C g \frac{ds_{C,\text{inst}}}{dt}$$

by using the derivative $\frac{d}{dt}$ of the instantaneous contractile stroke $s_{C,\text{inst}}$ (numerical time derivative on stroke time series data). For obtaining the instantaneous power density ($P_{d,\text{inst}}$) in Fig. 1D (ii), we divide the instantaneous power by the system mass:

$$P_{d,\text{inst}}(t) = \frac{P_{\text{inst}}(t)}{m_s}$$

Performance characterization of thin McKibben actuators

We characterized three identical McKibben actuators with 1.1 mm ID, 1.5 mm OD, and 245 mm length (fig. S4C). We mounted each actuator vertically onto a rigid plate, fixed at the top, with the bottom side connected to a mass holder. Loads ranging from 0 g to 500 g were applied in 100 g increments. The load included a 50 g holder, such that the "0 g" condition corresponded to the holder being removed. The bottom side of the actuator was also connected to a displacement meter, identical to that used in the muscle pair characterization, while pressure was measured via a pressure sensor placed at the top connection. Each actuator was manually filled at atmospheric pressure using a syringe. After filling, the actuator was loaded, and pressure was applied at a constant flow rate by a syringe pump that injected 1 mL of Novec fluid over 30 seconds. Data is acquired with the same DAQ device used in the muscle pair characterization tests at a sample rate of 10/s.

Pressure-flow characterization of fiber pumps

Fresh dielectric liquid is used for each test. Each test was performed on $n = 3$ pumps under three different pressure conditions: $p_{\text{atm}} = 0$ kPa (open circuit), $p_{\text{bias}} = 0$ kPa (closed circuit) and $p_{\text{bias}} = 75$ kPa (closed circuit). We measured both maximum flow rate (keeping the pump directly connected to the flow meter in a loop) and maximum pressure (by closing the 3-way valve between pump and flow sensor) (movie S3). We kept the two McKibben actuators in the system during fiber pumps' characterization for two reasons: (1) using the same configuration of fluidic circuit used to operate the electrofluidic muscles, and (2) to smoothen the pressure drops due to viscoelastic creep of the plastic components in the circuit. If the fiber pump is only connected to tubing and valve, the fluidic circuit becomes very sensitive to small mechanical deformations of its components when pressure increases, since small changes in volumes lead to significant changes in pressure. The presence of McKibbens as small pressure vessels helped in smoothening this response, so small volume changes due to viscoelastic deformations of the fluidic components became negligible in our measurements.

Pressure and flow rate data are recorded applying 1 kV voltage steps, from 1 to 8 kV. After a small transient of 1-2 seconds when increasing voltage, data are acquired for 10 seconds at each voltage step. The first series of voltage steps is discarded since it is used as conditioning for each pump. We observed that the pump performance became steady at fixed voltage after the first set of voltage steps from 1 to 8 kV. Furthermore, testing the pump under both open- and closed-circuit configurations, as well as across varying bias pressure conditions, revealed no measurable difference in performance at corresponding voltage levels (Fig. 2B ii and iii). However, under the $p_{\text{atm}} = 0$ kPa condition (open circuit), the maximum safe operating voltage reached was 7 kV. In contrast, under the $p_{\text{bias}} = 0$ kPa condition (closed circuit), the onset of cavitation reduced this threshold to 4 kV, as demonstrated in movie S2. Increasing the bias pressure to $p_{\text{bias}} = 75$ kPa (closed circuit) enabled stable pump operation up to 8 kV. Reported data for each voltage value are the average and standard deviation over 10 seconds of measurement and across $n = 3$ devices.

Determining cavitation thresholds

Cavitation thresholds were determined through controlled high-voltage experiments with stepwise voltage increments. The experiments were repeated for 0, 25, and 50 kPa bias pressure conditions (Fig. 3F, iv). A 5 M Ω resistor was placed in series with the pump to

limit current, enabling continued data acquisition even in the event of dielectric breakdown. Voltage was supplied by a custom high-voltage power supply in fig. S8A, comprising an Arduino Nano Every microcontroller and a DC converter (Advanced Energy ULTRAVOLT 10A series, 15 W maximum output power, 1.5 mA maximum output current). During testing, inlet and outlet pressures, contraction and extension strokes, applied voltage, and current were continuously recorded using two Digilent Discovery MCC-234 data acquisition (DAQ) devices operating in differential mode at a sampling rate of 500 Hz. The two DAQ units were synchronized via a shared clock.

Voltage was applied in 1 kV increments, starting from 2 kV, and maintained for 20 seconds at each step. For each voltage level, five independent repetitions were conducted. Within each repetition, data were averaged over a 10-second window, and the resulting values were further averaged across the five repetitions to obtain a representative measurement. This procedure was repeated for each bias pressure condition.

Movie S2 shows live measurements of dielectric breakdown in the muscle under a 200 g load at three different bias pressure conditions: 0 kPa, 25 kPa, and 50 kPa. Dielectric failure occurs at 5 kV for $p_{\text{bias}} = 0$ kPa, 6 kV for $p_{\text{bias}} = 25$ kPa, and 7 kV for $p_{\text{bias}} = 50$ kPa, indicating an increase in the breakdown threshold with higher bias pressure.

Characterization of individual muscle pairs (one fiber pump connected to two McKibben actuators)

All the fiber pumps used in this work have the same design and follow the fabrication method reported above, they feature an ID of 1.1 mm, OD of 1.6 mm, L of 350 mm, and bare platinum as the metal wire. The McKibben actuators used in the muscle pair characterization experiments are 280 mm long and fabricated with the method outlined above, characterized by an inner TPU tubing ($ID = 1.1$ mm, $OD = 1.5$ mm) and outer nylon braided sleeve ($ID = 1.6$ mm). The overall weight of the system varies according to the applied bias pressure due to the additional volume of fluid in the system as reported in table S2.

Before each test, the system was filled with Novec 7100 dielectric fluid using two sterilized polypropylene syringes on both ends of the fluidic circuit. The syringes also helped in removing air bubbles trapped in the system to avoid electrical breakdown. After filling, we applied extra liquid volume which increased the bias pressure of the system. When the desired bias pressure value was reached, the circuit was closed with 3-way fluidic valves. Loads are added as the last step before voltage application and starting the experiment.

a. Quasi-static response: Varying voltage, varying load and varying bias pressure experiments

We conducted cyclic tests to characterize the performance of individual muscle pairs at different values of applied bias pressure, voltage, and load. Each test was performed on three identical devices, each device comprising one fiber pump and two McKibben actuators and repeated over five cycles at a frequency of 0.05 Hz. For each voltage value, the power supply output was adjusted, and the pump response was allowed to stabilize for 30 seconds at constant voltage before starting cyclic actuation at 0.05 Hz. Each data point represents

the average over 10 seconds per cycle, five cycles per device, further averaged across the three devices.

The first set of tests analyzed the response of individual muscle pairs under a constant load of 200 g with constant voltage applied in 0.05 Hz cycles for 100 seconds. Each test was repeated varying the applied voltage from 4 to 8 kV (1 kV increments) and for different values of bias pressure from 0 kPa to 125 kPa, in 25 kPa increments.

The second set of tests were conducted by applying a fixed, constant voltage of 6 kV in all the tests while varying the symmetric load (both actuators loaded equally) from the preload value (25 g) up to 500 g, in 100 g increments. Each set of tests was repeated for different bias pressure levels ranging from 0 to 125 kPa, in 25 kPa increments. The second set of tests were repeated also in the asymmetric loading condition (only contracting actuator loaded) at constant voltage of 6 kV, using the same protocol of the symmetric loading experiment.

b. Dynamic response: Varying frequency and varying load experiments

The dynamic response of the muscles was characterized on $n = 3$ devices. Each test followed our standardized protocol: (i) application of bias pressure, (ii) pressure stabilization period, (iii) application of mechanical load, (iv) voltage application at the target frequency for 20 seconds, (v) power-off, and (vi) load removal. Extracting the displacement sensor maximum and minimum values for every cycle required running a moving maximum and minimum operator over a timespan, which was five actuation cycles, before and after every timestep.

We characterized the dynamic response of electrofluidic muscle pairs by measuring contraction under varying loads, at actuation frequencies from 0.25 Hz to 10 Hz (unipolar square wave, 50% duty cycle). Voltage was fixed at 7 kV and bias pressure at 75 kPa (optimal value from quasi-static characterization). Muscles were symmetrically loaded with 100 to 400 g, in 100 g increments. The setup in fig. S6A is the same as used in quasi-static characterization. We monitored the endpoint of the contracting actuator and recorded contraction over time with a sampling frequency of 500 Hz for 20 s (fig. S6B). Muscle dynamic response is shown in movie S4.

Figure S6C shows the average maximum contraction over a 20-second interval for each frequency and load. Contraction remains nearly constant up to 1 Hz, then starts decreasing significantly—by $\geq 25\%$ of maximum value— from 3 Hz onward. This decrease is due to the time it takes for the pump to fill the actuators with liquid. Therefore, the bottleneck in dynamic response is the flow rate of fiber pumps, as confirmed by our model. To address this, we built a fast-response demonstrator, reported in the next section, using four fiber pumps in parallel to drive two McKibben actuators, dramatically decreasing response time.

The gray-shaded area in fig. S6C highlights frequencies where non-periodic oscillations occurred (multiple peaks per cycle or chaotic response). Such response is due to the strongly nonlinear behavior of McKibben actuators when they cross zero tension. Being fibers, they buckle under compression and their stiffness changes dramatically compared to when under tension by the masses. This nonlinear response hides the resonance peak, which starts to appear at 4 Hz for 100 g and 3 Hz for 400 g. In the unshaded frequency range, the actuators did not cross zero tension, so the response remained periodic. Examples of non-periodic oscillations are shown in fig. S7.

We report nonperiodic muscle behavior within the frequency range of 3 – 7 Hz (fig. S7). Increasing the applied load shifted this range toward lower frequencies: under a 100 g load, nonperiodic oscillations were observed between 4 – 7 Hz, whereas under a 400 g load, it occurred between 3 – 6 Hz. Representative examples of muscle contractions deviating from the applied square wave voltage are shown in fig. S7.

c. Long-term stability experiments

Long-term stability tests were conducted under identical conditions using the setup shown in Fig. 3A. All data—including inlet and outlet pressures, contraction and extension strokes, applied voltage, and current—were recorded using the same DAQ setup at a sampling rate of 500 Hz. To evaluate the stability of an individual muscle pair, we applied a 2 Hz square wave at 50% duty cycle for 5000 cycles (fig. S6D). A bias pressure of 75 kPa was used, and the McKibben actuators were symmetrically loaded with 200 g loads. The driving voltage was set to 7 kV, corresponding to ~90% of the maximum voltage applied in the main experiments. The muscles showed excellent stability in terms of contraction, pressure difference, and electrical current.

d. Statistical analysis

For all muscle pair characterizations, we repeated each test on three identical devices, each made of one fiber pump and two McKibben actuators. Each test consisted of five cycles unless reported otherwise in the figure captions. Each robotic application characterization consisted of a single device and 10 cycles for each data point, reporting the average and standard deviation over the total number of repetitions.

The data analysis for muscle pairs experiments and robotic applications has been conducted using Python scripts. The original sampling rate was reduced to 50 Hz for computational enhancement without any appreciable difference in statistical outcomes. All data are unfiltered unless reported otherwise in figure captions. The analysis of cyclic experiments required delimiting the duty cycle of every cycle from the raw time series. Feed voltage and output current varied more sharply than the mechanical measurements of pressure and displacement. Therefore, we identified the starting and end times of every duty cycle based on electrical measurements and adopted those times for evaluating variations in pressure and displacement.

Demonstrations with modular muscle architectures

Fabrication and assembly of lever arm demonstration

The lever arm assembly consisted of an antagonistic pair of thin McKibben actuators ($ID = 1.1$ mm, $OD = 1.5$ mm, $L = 340$ mm) (Fig. 4A). Four identical fiber pumps ($ID = 1.15$ mm, $L = 350$ mm), fabricated using platinum electrode wires, were used to drive the actuators. The muscle with four parallel pumps weighs 8 g as filled. Movie S5 demonstrates the full assembly procedure of lever arm demonstration.

To achieve uniform pressure distribution and sealed connections, two custom-designed 3D-printed manifolds (fig. S9 and movie S5) were used to fluidically connect the four fiber pumps in parallel, followed by series connection with the McKibben actuators. Stainless steel needles were fixed into the manifolds with UV-curable epoxy to interface with 95A

TPU tubing at both ends of the McKibben actuators. Four additional needles were installed at the opposite end of each manifold to establish fluidic connections to the fiber pumps (fig. S9 and movie S5).

First, the fiber pumps are assembled around a 3D printed bobbin attached to the vertical frame. Then, the McKibbens were integrated into the lever arm structure, consisting of a 30 cm acrylic beam equipped with a mechanical stopper at the horizontal (180°) position. The agonist actuator was attached to the arm, 15 mm from the pivot point, and the antagonist actuator was attached 20 mm from the pivot. The antagonist was assembled in a slack state to prevent any braking effect during contraction of the agonist.

Force and strain rate characterization for the lever arm muscle

The lever arm static force is calculated (fig. S10A), by balancing the torques around the arm joint, as

$$F = \frac{d_W m_W + d_F m_F}{d_A} g,$$

where α is the tracked absolute angle between the arm and the vertical at the ball takeoff time, distances and mass values are specified in fig. S10A, and g is the gravitational acceleration.

The actuator strain was tracked using the video tracking software (Kinovea) and the strain rate (% s⁻¹) computed as

$$\text{strain rate} = \frac{\text{strain}}{t}$$

where the lever arm strain and time refer to the ball take-off instant, computed at steady state.

Fabrication of Electrofluidic Muscle Bundle pair

The muscle bundle pair demonstration is composed of two 8-McKibben actuator bundles, with four fiber pumps in parallel-series configuration. The first muscle is composed of two fiber pumps connected in parallel wound around an 8-McKibben actuator bundle in a circular arrangement. The second muscle is assembled following the same design. Finally, the two parallel pump pairs are connected fluidically in series resulting in an antagonistic muscle bundle pair. Movie S7 demonstrates the full fabrication procedure of the muscle bundle pair.

Each muscle bundle is made of eight thin McKibben actuators (Fig. 5A) sharing the same fluidic input and output ($ID = 1.25$ mm, $OD = 1.75$ mm, $L = 280$ mm) and two fiber pumps ($ID = 1.1$ mm, $L = 350$ mm). To ensure a uniform distribution of the pressure inside the bundle and a sealed connection, two 3D printed cylindrical manifolds were designed to fluidically connect eight McKibben actuators and two pumps together. Stainless steel needles were fixed in place using UV-epoxy into the fluidic manifolds to establish the connection with the 95A TPU tubing at the two ends of the McKibbens. Lateral needles were added to the manifold to establish fluidic connection with the two fiber pumps (fig. S9

and movie S7). Each muscle bundle has a length of 280 mm, with a diameter of 7 mm and an overall weight of 11 g, as filled at 135 kPa bias pressure.

Performance characterization of Electrofluidic Muscle Bundle pair

An experimental setup (Fig. 5B) was built to test the muscle bundle pair. Two pressure sensors were connected to the antagonist and agonist muscle bundles, respectively monitoring the inlet and outlet pressures generated by the pumps when activated. The muscles were vertically hung on the setup frame using a 3D printed support that holds the muscle in place. At the lower ends of each muscle, we attached robust 3D printed load holders (for the application of various high loads) which were then connected to two displacement sensors to measure agonist contraction and antagonist elongation during operation. During the tests, all the data (inlet and outlet pressures, contraction and elongation, applied voltage, current) were measured and recorded using the same dual-DAQ setup.

Prior to testing, the system was filled with dielectric fluid using two polypropylene syringes, after which additional fluid was added to reach a bias pressure of 135 kPa. All the 3-way valve outlets were closed except for the two outlets connected to the two pressure sensors for monitoring inlet and outlet pressures during the tests. For varying load tests, before adjusting the power supply output, muscles were loaded which increased the initial pressure of the system (similar to the individual muscle pair tests in Fig. 2E)

For evaluating the performance of muscle bundles, three separate tests were carried out. The first set of tests was conducted by applying a fixed preload to each muscle and analyzing the performance of the system at a voltage range of 4 - 7 kV (Fig. 5E, i). The preload corresponds to the mass of the 3D printed load holder which is 250 g. The second set of tests was conducted by varying the value of symmetrically applied loads from 500 g – 3 kg (in increments of 500 g) to each muscle bundle, at a voltage step of 7 kV (Fig. 5E, ii). The third set of tests was conducted by loading the contracting muscle only (asymmetric loading) from 1 – 4 kg, in increments of 1 kg, at 7 kV voltage step (Fig. 5E, iii).

For each test, the system was actuated for 10 cycles with a duration of 10 seconds each and data were recorded at a sample rate of 500 Hz. Once the power supply is turned off, the loads are removed from the muscles to allow the system to go back to the initial position.

Fabrication of Electrofluidic Woven Muscle pair demonstration

The woven muscle pair demonstration is composed of two 10-McKibben flat actuators, with four fiber pumps in parallel-series configuration. The first muscle (biceps) is composed of two fiber pumps connected in parallel intertwined with a 10-McKibben flat actuator. The second muscle (triceps) is assembled following the same design. Finally, the two parallel pump pairs are connected fluidically in series resulting in an antagonistic woven muscle pair. Movie S9 demonstrates the full fabrication procedure of the woven muscle pair.

Each woven muscle is composed of ten thin McKibben actuators (Fig. 6A) sharing the same fluidic input and output ($ID = 0.7$ mm, $OD = 1$ mm, $L = 420$ mm) and two fiber pumps ($ID = 1.1$ mm, $L = 350$ mm). To ensure a uniform distribution of the pressure inside the flat muscle and a sealed connection, two 3D printed rectangular manifolds were designed to fluidically connect ten McKibben actuators and two pumps together. Stainless steel needles

were fixed in place using UV-epoxy into the fluidic manifolds to establish the connection with the 95A TPU tubing at the two ends of the McKibbens. Lateral needles were added to the manifold to establish fluidic connection with the two fiber pumps (fig. S9 and movie S9). Weaving the McKibben actuators with the pumps resulted in an approximate 20 mm reduction in the woven muscle length (from an initial McKibben length of 420 mm), due to out-of-plane travel imposed by the weave structure. Each woven muscle has a final length of 400 mm, with a width of 60 mm and an overall weight of 18 g, as filled at 120 kPa bias pressure.

Performance characterization of Electrofluidic Woven Muscle pair

An experimental setup (Fig. 6B) consisting of a fully 3D printed human-scale arm was built to characterize the woven muscle pair performance. The arm is constructed of an upper arm connected to the forearm with a 3D printed flexible joint (Formlabs, Flexible 80A), after which the forearm is press-fitted a 3D printed hand model with a mass attachment point. The top section of each muscle was attached to a 3D-printed rigid peg block that allowed us to adjust the upper attachment point. The lower ends of each muscle were attached to 3D printed flexible arm straps that wrap around the forearm. Similarly, the lower attachment point of each muscle can be simply adjusted. The triceps muscle was attached at a lower point than the biceps muscle at their top sections, leaving it slack, to reduce the amount of braking effect during actuation cycles.

For data measurement, two pressure sensors were connected to the triceps (antagonist) and biceps (agonist) muscles, respectively monitoring the inlet and outlet pressures generated by the pumps when activated. Applied voltage was monitored using a high voltage probe and electrical current was measured during operation, using the same dual-DAQ setup. The position of the forearm and the bending angle were tracked using the video processing software (Kinovea).

Prior to testing, the system was filled with dielectric fluid using two polypropylene syringes, after which additional fluid was added to reach a bias pressure of 120 kPa. All the 3-way valve outlets were closed except for the two outlets connected to the two pressure sensors for monitoring inlet and outlet pressures during the tests. For varying load tests, before adjusting the power supply output muscles were loaded which increased the initial pressure of the system (similar to the bundle and individual muscle pair tests).

For evaluating the performance of woven muscles, the tests were carried out in three sets. The first set of tests consisted of lifting only the forearm (preload = 350 g) and analyzing the performance of the system at a voltage range of 4 – 8 kV, in 1 kV increments (Fig. 6D). The second set of tests was conducted by varying the applied load (0 – 600g, in increments of 200 g) at an 8 kV voltage step (Fig. 6E). The load was hung on the model hand with a Kevlar thread. The third set of tests were conducted to evaluate the how varying the number of McKibben actuators in the biceps muscle affects the muscle's response time (Fig. 6F). Since the triceps muscle acts as a reservoir and a non-restricting muscle in this assembly, it doesn't have any effect on the muscle pair's response time. Therefore, we only decreased the number of McKibben actuators in the biceps muscle, from 10 actuators down to 2, removing 2 actuators each time. The McKibbens were manually removed from the manifold and replaced by short, shore 95A TPU tubing sections sealed with 3D printed barbed caps to ensure a closed fluidic circuit.

For each test, the arm was actuated for 10 cycles with a duration of 10 seconds each and data were recorded at a sample rate of 500 Hz. Once the power supply is turned off, any added load is removed and the arm is allowed to go back to the initial position.

Force and strain rate characterization for Electrofluidic Woven Muscle

The woven muscle static force is calculated (fig. S10B), by balancing the torques around the arm joint, as

$$F = \frac{\sum_{i=H,W,F} m_i d_i}{d_a + \frac{d_b}{\tan \alpha}} g,$$

where α is the tracked absolute angle between the forearm and the vertical axis at maximum arm flexion, distances and mass values are specified in fig. S10B, and g is the gravitational acceleration. The denominator takes into account that the muscle is attached to the model arm far from its cross-section center of mass (fig. S10B).

The strain was tracked using the video tracking software (Kinovea) and the strain rate (% s⁻¹) computed as

$$\text{strain rate} = \frac{\text{strain}}{t_{90}}$$

where the response time of the woven sheet muscle is computed at 90% of the maximum strain.

Mechanical model of the electrofluidic fiber muscle

The thin McKibben actuators fabricated for this research consist of slender elastomeric tubes composed of thermoplastic polyurethane (TPU), externally constrained by a braided nylon sheath. These actuators undergo deformation when pressurized with a working fluid, in this case a dielectric liquid (Novec 7100). The TPU tube is modeled as a membrane undergoing axisymmetric deformation, and the fluid is assumed to be incompressible. The mechanical behavior of the TPU is described using an incompressible Neo-Hookean hyperelastic strain energy function, which quantifies the elastic energy stored in the membrane as a function of strain. The nylon braid is modeled as an inextensibility constraint along the direction of its filaments. Actuator deformation—and corresponding changes in stored elastic energy—occurs either through internal fluid pressurization or external mechanical loading.

With the mechanical behavior of individual actuators characterized, the quasi-static coupling behavior of the full system can be analyzed. Specifically, the system is modeled under the assumptions that: the total volume of bias fluid is conserved; at steady state, the pressure difference between actuators equals the pump's differential pressure under zero flow conditions; the flow is laminar, so that, at given EHD pump voltage, the characteristic pump pressure vs flow rate relation is linear. Given the actuator geometry, tubing elastic modulus, maximum pump pressure differential and flow rate, and applied end-effector force, the model enables prediction of several key performance metrics: steady-state end-effector displacement at a given bias pressure, the bias pressure that maximizes displacement, system response time, and, critically, the operational regime that avoids cavitation and subsequent electrical breakdown.

A schematic of the main model ingredients, i.e. elasticity, braiding inextensibility, flow assumptions and continuity, is depicted in fig. S1. Finally, the model is extended to consider various arrangements of EHD pumps and actuators, with the goal of assisting engineering modular muscle architectures.

Thin McKibben actuator model

A McKibben actuator is built by enveloping a soft elastomeric tube by a polymeric braiding. Its actuation relies on the quantity of fluid which flows in and out of it. Our McKibbens are able to produce important strokes, which amount up to about 30% of the actuator undeformed length. For this reason, we adopt the formalism of finite strain theory and introduce the deformation gradient $\mathbf{F}$ and the right Cauchy-Green deformation tensor $\mathbf{C}$

$$\mathbf{F} = \mathbf{I} + \nabla \mathbf{u}, \quad (1)$$

$$\mathbf{C} = \mathbf{F}^T \mathbf{F}, \quad (2)$$

where $\mathbf{I}$ is the identity tensor and $\mathbf{u}$ is the displacement field. Our actuators are cylindrical, slender and thin walled. Therefore, we model them as indefinitely long cylindrical membranes, and model the deformation as axial symmetric. In this way we neglect the mechanical variables variation through the thickness and the border effects at the extremities. As a consequence of symmetry, the principal stretches are assumed to be the axial, circumferential and radial ones, here denoted as λ , λ_r and λ_h , respectively. These can be measured as

$$\lambda = \frac{l}{l_0}, \quad \lambda_r = \frac{r}{r_0}, \quad \lambda_h = \frac{h}{h_0}, \quad (3)$$

where l , r and h are the current actuator membrane axial length, radius and thickness and the subscript 0 refers to the reference configuration. Our nylon braiding is far stiffer than our TPU tubing. This means that, the extension of the braiding filaments can be neglected with respect to the tubing fibers in the same filament direction. Assuming, for simplicity, that the spacing in between filaments is small with respect to the tubing circumference and that, being a parallel arrangement of springs, the two components undergo the same displacement field, we can homogenize the double material structure by introducing the following internal kinematic constraint on the so-called fourth deformation invariant

$$I_4(\mathbf{C}, \mathbf{m}_0) = \lambda_{m_0}^2 = \mathbf{C} \mathbf{m}_0 \cdot \mathbf{m}_0 = 1. \quad (4)$$

This disallows the equivalent homogenous material to stretch along the braiding filament direction, which in the reference configuration is along $\mathbf{m}_0$ (53). If θ_0 is the initial angle between the filaments and the axial direction, the internal constraint becomes

$$\lambda^2 \cos^2 \theta_0 + \lambda_r^2 \sin^2 \theta_0 = 1, \quad (5)$$

since $\mathbf{C} = \text{diag}(\lambda^2, \lambda_r^2, \lambda_h^2)$ and $\mathbf{m}_0 = [\cos \theta_0, \sin \theta_0, 0]$ (fig. S1C). Observe that this constraint is unchanged under mirroring across the actuator longitudinal direction, so that Eq. 5 accounts for both the families of braiding filaments. Furthermore, the equation above, which describes an ellipse of semi-axes $1/\cos \theta_0$ and $1/\sin \theta_0$, represents an upper bound to the actuator stretch, since $\lambda < 1/\cos \theta_0$, and allows to uniquely relate axial and radial stretches. The fluid volume contained in the actuator can be calculated from the radial and axial strain as

$$V = \lambda \lambda_r^2 V_0^f = \varphi V_0^f, \quad (6)$$

where φ is the liquid filling, which is one in the reference configuration, and $V_0^f = \pi r_0^2 L_0$ is the reference fluid volume, i.e. the volume of fluid contained in an unstretched actuator. By using Eq. 5, the filling can be written as function of the axial stretch only as

$$\varphi(\lambda) = \frac{1 - \lambda^2 \cos^2 \theta}{\sin^2 \theta} \lambda. \quad (7)$$

It is remarkable to notice that, due to the two kinematic constraints, i.e. elastomer incompressibility and braid inextensibility, this relation is independent on actuator loading, as it is experimentally validated by looking at figure (fig. S4C, ii). As the McKibben extension changes, the filling rate varies with axial stretch as

$$\varphi'(\lambda) = \frac{d\varphi}{d\lambda} = \frac{1 - 3\lambda^2 \cos^2 \theta}{\sin^2 \theta}. \quad (8)$$

If one evaluates this relation starting from the undeformed configuration, i.e. at $\lambda = 1$, it is easy to distinguish contractible- and extensible-type McKibben actuators, which respectively contract or extend with fluid injection, depending on the braiding angle as

$$\begin{cases} \theta_0 > \arccos 3^{-1/2} & \text{extensible} \\ \theta_0 < \arccos 3^{-1/2} & \text{contractible} \end{cases}, \quad (9)$$

where $\arccos 3^{-1/2} \approx 54.7^\circ$ is often named *magic angle* and appears ubiquitously in biological anisotropic tissues (54). In soft fibrous actuation, one is interested in contractible fiber muscles, since extendible ones would immediately buckle while pushing their end effector. In this case, the filling rate must be positive in contraction. This identifies the minimum theoretical stretch, corresponding to the maximum theoretical contraction, and the minimum theoretical stretch, corresponding to zero fluid volume. Summarizing, the range of possible stretches and fillings results

$$\text{contractile strain} = (1 - \lambda) \in (1 - \sqrt{3}\lambda_l, 1 - \lambda_l), \quad (10)$$

$$\varphi \in \left(0, \varphi_l = \frac{2\lambda_l^3}{3\lambda_l^2 - 1}\right), \quad (11)$$

$$\text{where } \lambda_l = \frac{1}{\sqrt{3}\cos\theta_0}. \quad (12)$$

For example, the maximum contraction is about 42%, 19% and 0% for braiding angles of 0° , 45° and the magic angle, respectively. It is therefore common practice to design contractible McKibben actuators with as low θ_0 as possible (55).

The internal energy of the actuator

$$U = \underbrace{w(\lambda)V_0^s}_{\text{strain energy}} - \underbrace{(p\varphi(\lambda)V_0^f + F\lambda L_0)}_{\text{external work}} + \text{constant} \quad (13)$$

accounts for:

- the tubing energy $V_0^s w$, where $V_0^s = 2\pi r_0 h_0 L_0$ and w are the unstretched tubing membrane solid volume and its energy density.
- the work done by the fluid pressure p for increasing the reference liquid volume V_0^f of $\varphi - 1$.
- the work of the end effector force F . A comprehensive review highlighting the importance of the tubing elastic energy for effective models can be found in (55). The total energy above aligns with Gaylord formula for his patented fluidic actuator (56, 57) if the strain energy is neglected.

At equilibrium, this internal energy must reach a minimum, so that

$$\frac{dU}{d\lambda} = 0. \quad (14)$$

This provides the McKibben output traction F as a function of pressure and stretch as

$$F = \frac{V_0^s w'(\lambda) - V_0^f \varphi'(\lambda)p}{L_0} = \pi r_0 (2h_0 w'(\lambda) - r_0 \varphi'(\lambda)p), \quad (15)$$

if pressure and stretch are known. In our experiments, though, force is imposed and pressure is generated by fluid inflow. At first, we pressurize the system at zero force by injecting some bias volume $\varphi_* = \varphi(\lambda_*)$ which results in a bias stretch λ_* until we reach the bias pressure

$$p_* = \frac{V_0^s w'(\lambda_*)}{V_0^f \varphi'(\lambda_*)} = \frac{2h_0 w'(\lambda_*)}{\underbrace{r_0 \varphi'(\lambda_*)}_{\text{bias pressure}}} \quad (16)$$

At this point we add a force F , but both the stretch stays constant since no fluid enters the actuator. The external work is therefore carried out by an isochoric increase in pressure, so that the initial pressure, before the pump is switched on

$$p_i = p_* - \frac{Mg}{\underbrace{\pi r_0^2 \varphi'_*}_{\text{loading pressure}}} \quad (17)$$

and the loading pressure equilibrates the applied force on the stretched area. Finally, at fixed force we let new fluid in and out of the McKibben and can model the pressure stretch relation as

$$p = \frac{V_0^s w'(\lambda)}{V_0^f \varphi'(\lambda)} - \frac{FL_0}{V_0^f \varphi'(\lambda)}, \quad (18)$$

where the first term depends on the soft tubing elasticity and the second one on the applied force. The common denominator embodies the braiding constraint. Notice that if we do not account for the energy expended for deforming the tubing, free extensions and contractions would be carried with zero force.

Since our tubing is a soft elastomer, out of simplicity, we adopt the standard incompressible Neo-Hookean energy density

$$w(\lambda) = \frac{\mu}{2} (\lambda^2 + \lambda_r^2(\lambda) + \lambda_h^2(\lambda) - 3), \quad (19)$$

where the relation between the circumferential and radial stretches with the axial one can be retrieved explicitly by use of Eq. 5 with the incompressibility constraint $\lambda \lambda_r \lambda_h = 1$. After some manipulation, it is also possible to write the McKibben strain energy derivative as

$$w'(\lambda) = \mu \lambda \left(1 - \tan^{-2} \theta_0 + \frac{2 \cos^2 \theta_0 \lambda^2 - 1}{(\lambda^2 - \cos^2 \theta_0 \lambda^4)^2} \sin^2 \theta_0 \right) \quad (20)$$

which is used to compute pressure. For contractible McKibben actuators, both w' and φ' are negative, i.e. the stored elastic energy and the filling volume increase with contraction.

System model for muscle pairs

Our fiber muscle pair is composed by two bundles of identical antagonist McKibben actuators, connected by a modular system of fiber pumps. This connection imposes an equal pressure on the two sides and sets a pressure difference when the pumps are switched on. This muscle is controlled by the voltage supplied to the pumps electrodes and is able to exert forces for moving its end effectors. For simplicity, let us consider the simplest coupled system, with two identical actuators, lifting the same weight. The fiber pump is expected to provide a maximum pressure difference $P_{\max}$ at zero flow and a maximum flow rate at zero pressure $Q_{\max}$ (Fig. 1B). If the pump works in laminar regime, the flow vs pressure relation is linear (fig. S1D)

$$Q = Q_{\max} - \frac{Q_{\max}}{P_{\max}} P, \quad (21)$$

where $P_{\max}$ and $Q_{\max}$ depend on the applied voltage for a given pump (40). The pressure difference between the two actuators is

$$P(t) = p(\lambda(t)) - p(\tilde{\lambda}(t)), \quad (22)$$

where with the symbol $\sim$ we identify the antagonistic actuator. The flow rate, seen by the actuators, is the same in absolute value, so that

$$Q(t) = V_0^f \dot{\phi} = -V_0^f \dot{\tilde{\phi}} \quad (23)$$

$$\underbrace{\varphi(\lambda) + \varphi(\tilde{\lambda}) = 2\varphi^*}_{\text{mass conservation}} \rightarrow \tilde{\lambda} = \tilde{\lambda}(\lambda) \quad (24)$$

which preserves the initial fluid volume, i.e. the whole fluidic muscle bias filling $2\varphi_*$, at any time.

For modelling the coupled system time response, it is necessary to integrate the pump evolution nonlinear ODE Eq. 21, together with the coupling conditions for pressure and flow rate and the initial condition, i.e. the biased configuration. Rather, we characterize the ODE by studying the system response under step voltage application. At zero time, the system is in its biased configuration and suddenly a certain voltage is applied to the fiber pump. Since the ODE is damped by the laminar fluid flow, it is expected to reach a constant steady state, where the flow rate is zero and the difference in pressure between the two actuators is therefore $P_{\max}(\text{voltage})$.

In formulas, the steady state condition is

$$p(\lambda_\infty) - p(\tilde{\lambda}_\infty) = P_{\max}, \quad t \rightarrow +\infty, \quad (25)$$

which we numerically solve together with the mass conservation Eq. 24, given a bias configuration and the pump $P_{\max}$ at given voltage.

This procedure has been followed for retrieving the steady state contractions vs bias pressure in Fig. 3E, where the plotted strain is $\varepsilon_\infty = \lambda_* - \lambda_\infty$, i.e. the actuator stretches relative to the biased configuration. Besides, one can compute the inlet pressure at steady state $\tilde{p}_\infty$ via Eq. 18 (as in fig. S3C). In the same Fig. 3E, the dotted lines identify the region where $\tilde{p}_\infty < 0$, i.e. approximately where the breakdown arises. The values of $P_{\max}(\text{voltage})$ are obtained by the pump characterization at zero flow (fig. S1D and Fig. 1B). The shear modulus has been identified by fitting the pressure model Eq. 18 to the single actuator pressure response (fig. S4C, i). The obtained value $\mu = 0.48$ MPa agrees well with the well known Gent formula for low hardness materials (58) which would return 0.56 MPa for incompressible Neo-Hookean solids with Shore A hardness 40. The braiding angle was 17° by construction in all our hand made McKibbens. The actuator length was $L_0 = 245$ mm. The tubing had an $OD = 1.501$ mm, $ID = 1.143$ mm, so that $r_0 = 0.661$ mm, $h_0 = 0.179$ mm.

On the other end of the step voltage process, i.e. for small times, the pump has not built pressure yet, so $P \approx 0$ and the system is in its bias configuration. The ODE Eq. 21 at zero time is

$$V_0^f \dot{\phi} = -V_0^f \dot{\tilde{\phi}} = Q_{\max}, \quad t \rightarrow 0, \quad (26)$$

which also identifies the flow characteristic time

$$\tau = \frac{V_0^f}{Q_{\max}}. \quad (27)$$

This is the time needed for injecting a unitary fluid volume at maximum pump flow rate, which in turn depends on voltage $Q_{\max} = Q_{\max}(\text{voltage})$ for a given pump. The McKibben has two limiting fillings (see Eq. 11), zero and φ_l . If the bias pressure is extremely high, the agonist McKibben will soon reach the steady state after very small contraction and its filling will arrive soon at its limit φ_l with the limiting speed $1/\tau$. If we search for a law relating response time to φ_* as $t_R = t_R(\varphi_*)$, this can be stated as

$$t_R(\varphi_* = \varphi_l) = 0, \quad (28)$$

$$\left. \frac{dt_R}{d\varphi_*} \right|_{\varphi_* = \varphi_l} = -\tau. \quad (29)$$

On the other hand, at zero bias filling, whereas the agonist McKibben has plenty of available contraction, the antagonist has none, because it cannot be further emptied. This writes

$$\left. \frac{dt_R}{d\varphi_*} \right|_{\varphi_* = 0} = \tau. \quad (30)$$

A very simple law for matching these two extremes is

$$t_R = \tau \varphi_* \frac{\varphi_l - \varphi_*}{\varphi_l} = \frac{V_0^f}{Q_{\max}} \varphi_* \frac{\varphi_l - \varphi_*}{\varphi_l}, \quad (31)$$

which predicts that the response time should show inverse proportionality to $Q_{\max}(\text{voltage})$ and that it's at its highest $\tau \varphi_l/4$ with a bias filling of $\varphi_l/2$. We recall that φ_l is uniquely determined by the braiding angle for incompressible tubing and infinitely stiff braiding filaments.

The response time vs bias pressure can be represented by mapping $\varphi_*(\lambda_*)$ Eq. 7 to $p_*(\lambda_*)$ Eq. 16 via the bias stretch λ_* as it is done for plotting Fig. 3F and fig. S2B.

Simplified model for preliminary design

Here we develop handy formulas for enabling a preliminary system design in view of possible applications. The Neo-Hookean energy density in Eq. 19, if the through-thickness stretch is neglected, becomes

$$w \approx \frac{\mu}{2} \left(\frac{1}{\sin^2 \theta_0} - 2 \right) (1 - \lambda^2) = \frac{\mu_\theta}{2} (1 - \lambda^2). \quad (32)$$

Notice that we introduce an effective elastic modulus $\mu_\theta = (\sin^{-2} \theta_0 - 2)$ incorporating both the tubing elasticity and the braiding constraint. For our $\theta_0 = 17^\circ$ McKibben actuators, this homogenized modulus is about ten times larger than the tubing-only one.

Since the flow characteristic time is τ , the flow rate can be modelled with an exponential decay $Q_{\max} \exp(-t/\tau)$ in step response. Consequently, the pump delta pressure as $P = P_{\max}(1 - \exp(-t/\tau))$, which indicates that the total amount of hydraulic work provided to the system from the pump $W^{\text{hydr}} = \int_0^\infty P Q dt \approx P_{\max} V_0^f / 2$. The initial time corresponds to the symmetric biased configuration. As $t \rightarrow \infty$, the hydraulic work must cause a variation in elastic and gravitational energy, which are conservative.

Further, we linearize the volume filling, so that the mass conservation becomes

$$\lambda_{\infty} + \tilde{\lambda}_{\infty} = 2\lambda_*. \quad (33)$$

This implies that, as a first approximation, the potential energy does not change in a symmetrically loaded muscle pair. Indeed, its variation is $(2\lambda_* - \lambda_{\infty} - \tilde{\lambda}_{\infty})F$, which is null for mass conservation.

The approximated elastic energy change, accounting for both agonist and antagonist muscles,

$$\Delta U^{\text{el}} = V_0^s \frac{\mu_{\theta}}{2} (\lambda_{\infty}^2 + \tilde{\lambda}_{\infty}^2 - 2\lambda_*^2) = P_{\text{max}} V_0^f / 2, \quad (34)$$

should equal the approximated pump hydraulic work. Furthermore, if the filling vs strain relation is linear, one can infer that the optimal bias strain is in between $\lambda_* = 1$ which leads to cavitation at zero load and $\lambda_* = \lambda_l$, which is the limiting theoretical absolute muscle strain. In this way, if P_{max} is high enough, the agonist and the antagonist will relatively contract and extend of $(1 - \lambda_l)/2$. This simplified reasoning leads to a rough estimate for the relative contractile strain at $(1 - \lambda_l)/2$. Summarizing, for the symmetric system, the steady state agonist relative contractile strain $\varepsilon_{\infty} = \lambda_* - \lambda_{\infty}$, can simply be estimated from the pump hydraulic work

$$\varepsilon_{\infty} \approx \sqrt{\frac{P_{\text{max}} r_0 / h_0}{2\mu \left(\frac{1}{\sin^2 \theta_0} - 2 \right)}} \leq \frac{1 - \lambda_l}{2}, \quad (35)$$

which shows how the maximum contraction achievable for a given muscle pair scales with the square root of the maximum pump pressure difference. Notice that, by removing the nonlinearity in the filling vs contraction relation, i.e. by neglecting the limiting contraction and filling, it is not possible to assess the influence of weight and bias pressure in a concise fashion. Nevertheless, we hold the above expression extremely valuable in providing rough estimates, as it is shown by comparing it with the experimental data (fig. S2G).

Modular muscle architectures

Our thin electrofluidic muscles are highly modular, since both pumps and actuators can be arranged in a versatile fashion in view of their robotic application, for achieving a wide range of forces and contraction rates (Fig. 1E). Basic pump arrangements are in series and parallel. The thin McKibbens are arranged in parallel, whereas their length can be modulated by cutting tubing and braiding in different sizes. The characterization of one pump module at varying input voltage corresponds to establishing the model relationship Eq. 21 which only has the experimentally measured p_{max} and q_{max} for that module. It has been shown in (40) that the pressure at zero flow is directly proportional to the pump length, or equivalently, we infer, to the number of pumps composing a series N_- . On the other hand, $N_{||}$ pump modules in parallel would provide a multiple of the flow at zero pressure, since we have shown in Fig. 2B that this quantity is unaffected by the environmental pressure both in open and closed circuit. Finally, a parallel arrangement of $N_{||}$ series of N_- modules, within the laminar flow assumption, modifies the overall architected pump behaviour as

$$Q = N_{||} q_{\text{max}} \left(1 - \frac{P}{N_- p_{\text{max}}} \right). \quad (36)$$

Let us consider the arrangement of N_{act} identical thin actuators per pump side, each exerting a force F/N_{act} . One actuator internal energy Eq. 13 becomes

$$U = N_{\text{act}} w(\lambda) V_0^s - \left(p \varphi(\lambda) N_{\text{act}} V_0^f + \frac{F}{N_{\text{act}}} \lambda L_0 \right), \quad (37)$$

to be minimized for obtaining the pressure stretch relation at given force. The flow characteristic time is now

$$\tau = \frac{N_{\text{act}} V_0^f}{N_{||} q_{\text{max}}}, \quad (38)$$

to be used for designing the response time in Eq. 31. It is clear that the response time, proportional to the characteristic time, grows with the number of actuators, i.e. the total amount of fluid to be pumped, and decreases with the number of pumps in parallel. This is the rationale for the muscle architecture designed in the lever arm application in Fig. 4.

The steady state pressure balance Eq. 25, to design the steady state contraction is also adapted by considering the average actuator force and $P_{\text{max}} = N_{-} p_{\text{max}}$. The simplified steady state contraction Eq. 35 now writes

$$\varepsilon_{\infty} \approx \sqrt{\frac{N_{-} p_{\text{max}} r_0 / h_0}{2\mu \left(\frac{1}{\sin^2 \theta_0} - 2 \right)}} \leq \frac{1 - \lambda_l}{2}, \quad (39)$$

which gives us the insight that steady state contractive strain grows with the square root of the number of pumps in series, i.e. of P_{max} , with the a priori kinematic constraint imposed by the braiding angle.

SUPPLEMENTARY FIGURES

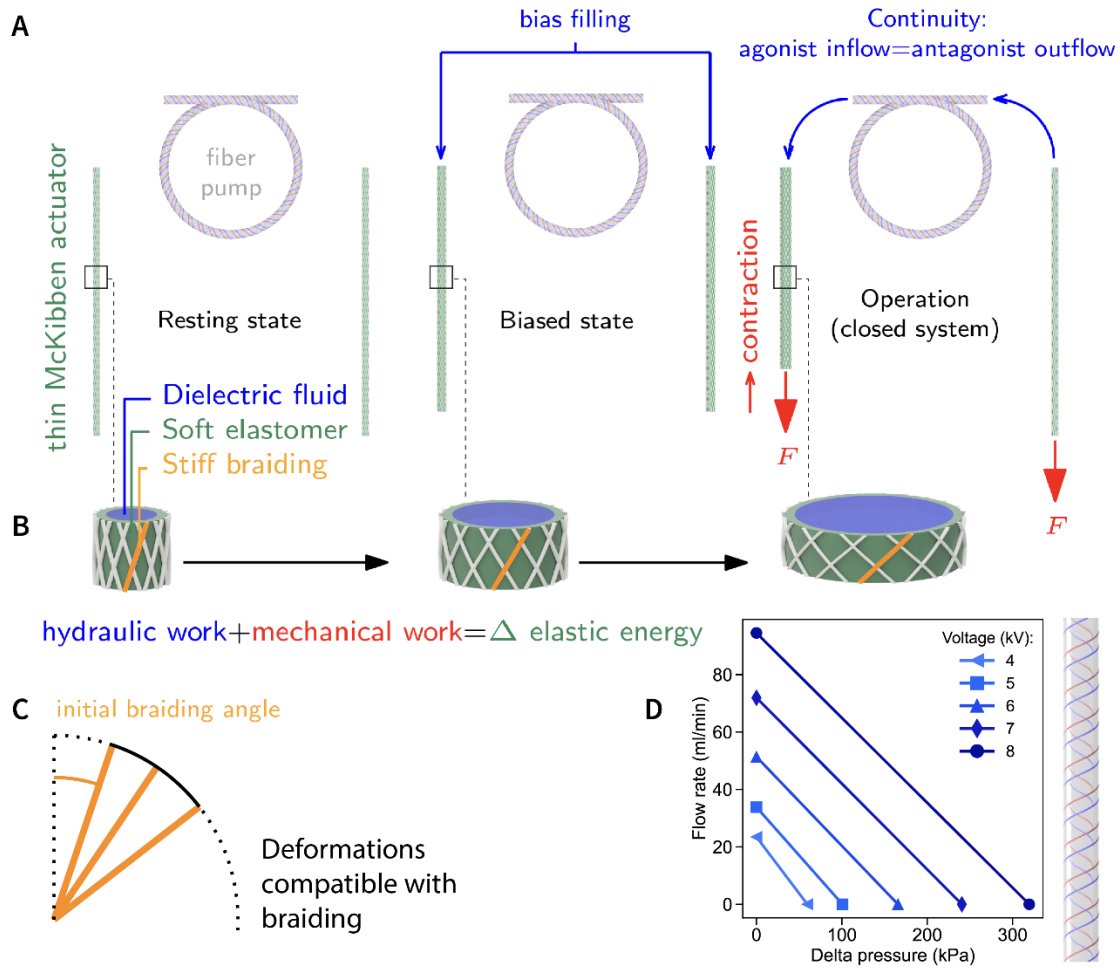


Fig. S1. Mechanical model of the electrofluidic fiber muscle. (A) The muscle is composed of thin McKibben actuators and fiber pump(s). At rest, no mechanical or hydraulic stress is present. The system is biased by liquid filling, which increases fluid pressure and uniformly strains the actuators. Once sealed, equal forces are applied while the pump is activated. The pressure differential between pump inlet and outlet, along with constant total fluid volume, enables agonist contraction. (B) Actuators are modelled as hyperelastic membranes kinematically superposed by stiff braiding, filled with incompressible dielectric fluid. Single actuator pressure is derived analytically, as a function of strain, by equating variations of membrane elastic energy to the amount of pressure hydraulic work and end effector mechanical work. (C) Kinematic collaboration between membrane and braiding is modelled by assuming rigid filaments, therefore restricting possible deformations. (D) Pump is assumed to operate in laminar flow, yielding a linear flow-pressure relationship that can be determined by experimentally measuring only the maximum flow at minimum pressure, and maximum pressure at zero flow (extracted from Fig. 2B). This, with the continuity equation, establishes a time-dependent coupling between agonist and antagonist actuators in the form of a time dependent nonlinear ordinary differential equation.

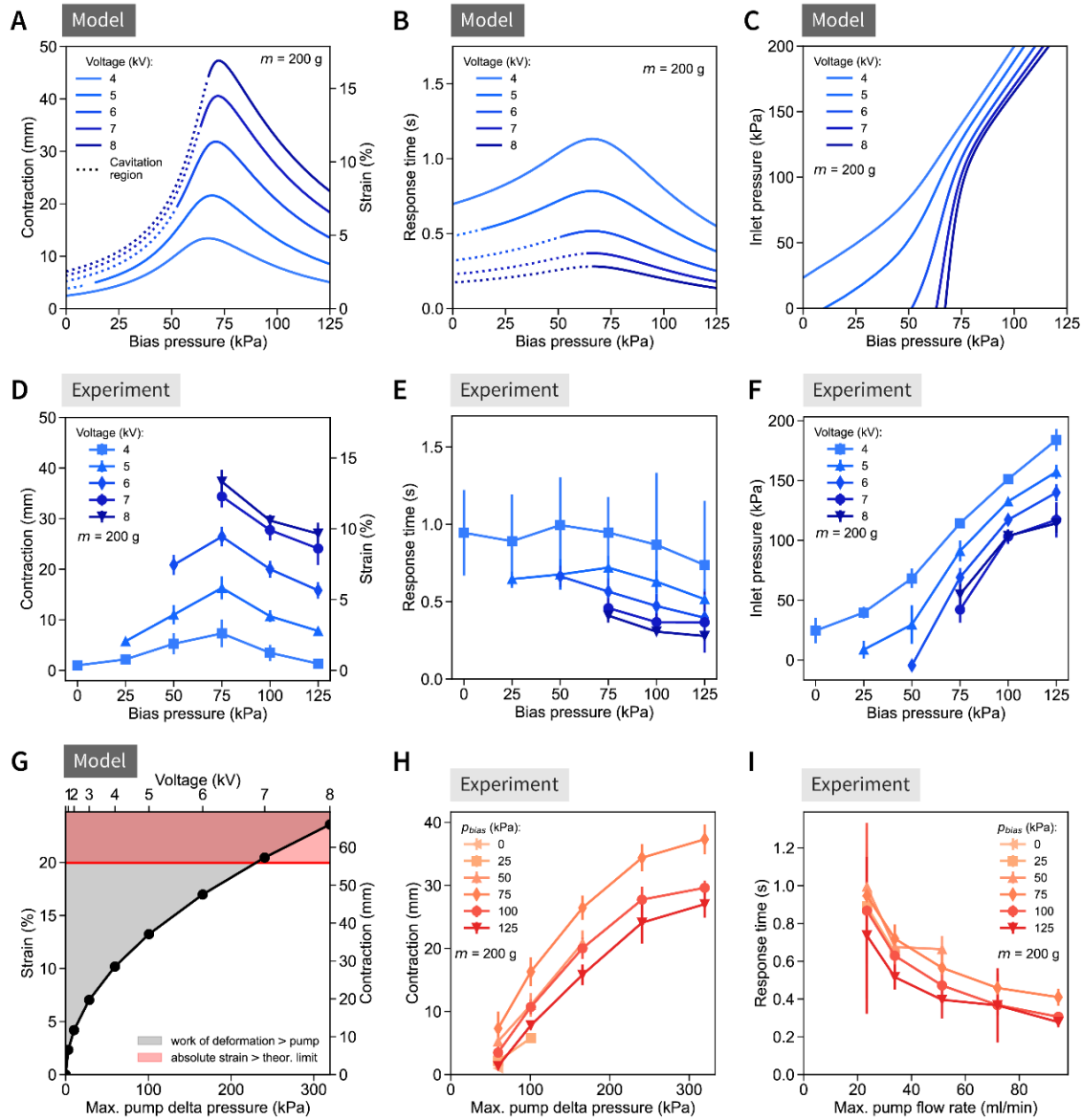


Fig. S2. Physical model of the individual symmetric muscle pair. Model results for varying voltage tests at a symmetrically applied fixed load of 200 g (Fig. 3). The model couples the McKibben actuator's nonlinear elasticity –arising from the interaction between the elastomeric tube and braiding– with the pump's experimentally measured maximum flow rate and maximum pressure for a range of applied voltage values (reported in Fig. 2B). The two model parameters are the measured braiding angle 17° and the soft tubing shear modulus 0.48 MPa, identified via McKibben actuator characterization tests in Fig. 2A. (A), (B) and (C) show the model predictions for the muscle response to a step voltage input. (D), (E) and (F) report the corresponding experimental results. (G) Region of achievable agonist strain based on simplified energy balance (white area), between hydraulic input energy and elastic energy, and mass conservation. (black) Pump pressure differential - at corresponding given voltage as experimentally determined - vs an upper bound to the actuator strain relative to the biased configuration at which the hydraulic work entirely transforms into elastic energy. A further upper bound (red) satisfies the constraint that the absolute actuator strain can at most reach its theoretical limit. (H) Experimentally achieved contraction as a function of pump maximum pressure. (I) Experimentally measured response time as a function of pump maximum flow rate. The inverse proportionality between response time and flow rate confirms the model findings. All values with error bars correspond to mean SD for $n = 3$ devices.

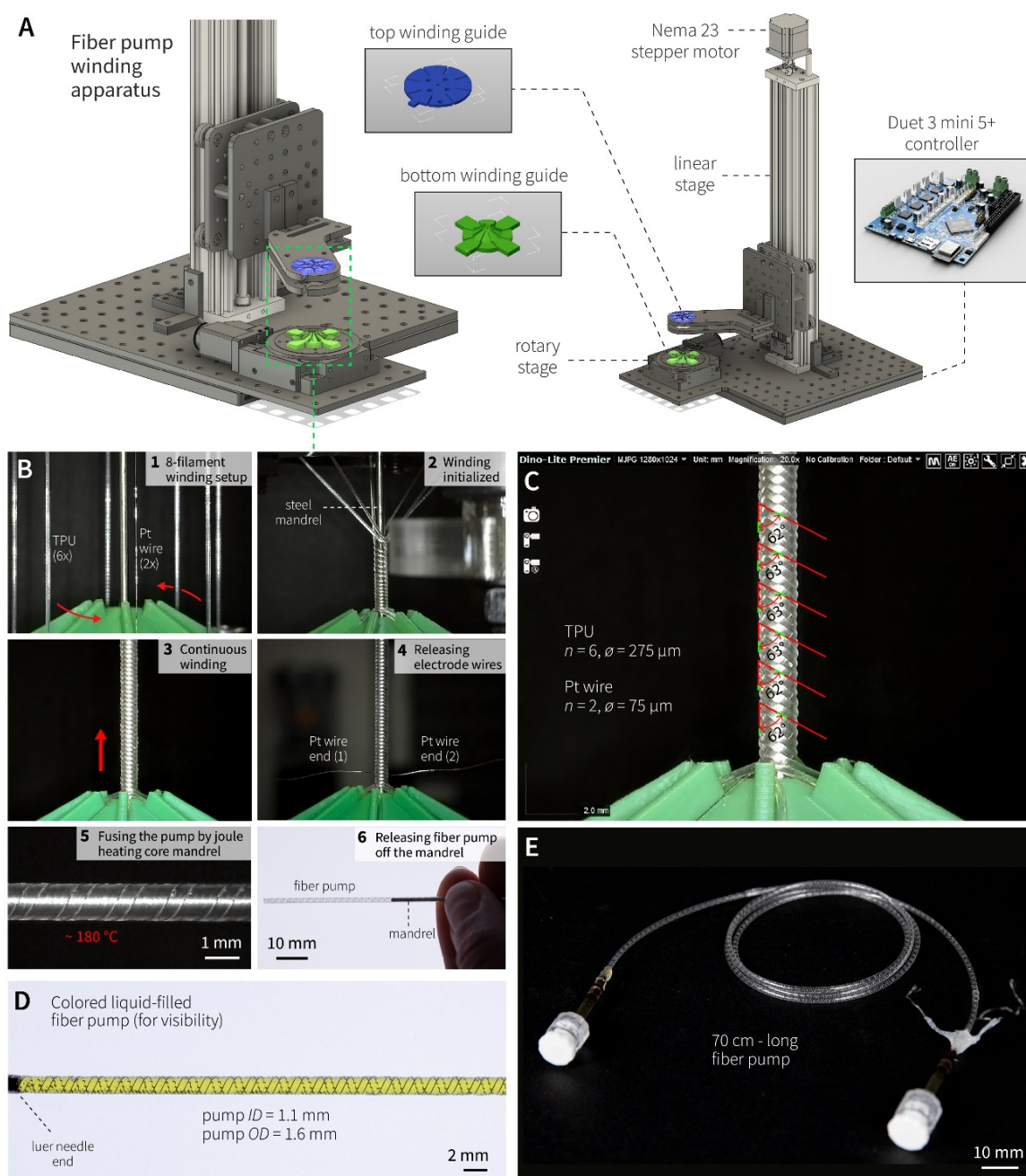


Fig. S3. Fiber pump fabrication by multifilament winding. (A) CAD model of the 8-filament winding apparatus. (B) The winding process used to fabricate fiber pumps. Six TPU filaments, two platinum (Pt) wires and a central stainless-steel rod is mounted (1). During winding, the Pt wire sits between the TPU filaments in an order imposed by the upper and lower winding guides (2, 3). The last step of the winding is to release the electrode wires at the two ends of the pump to allow access to the two pump terminals (4). Once winding is finalized, the mandrel is removed from the apparatus and the pump is fused by joule heating the core mandrel (5), which is subsequently removed (6). (C) Close-up image of the pump during the winding process. TPU filaments and wire electrodes are wound at a constant angle of $\sim 63^\circ$. (D) The fiber pump features an inner diameter (ID) of 1.1 mm and outer diameter (OD) of 1.6 mm. The pump is temporarily filled with a colored liquid to enhance the electrodes visibility. (E) Maximum pump length that can be fabricated on our winding apparatus. This length was not used in the article.

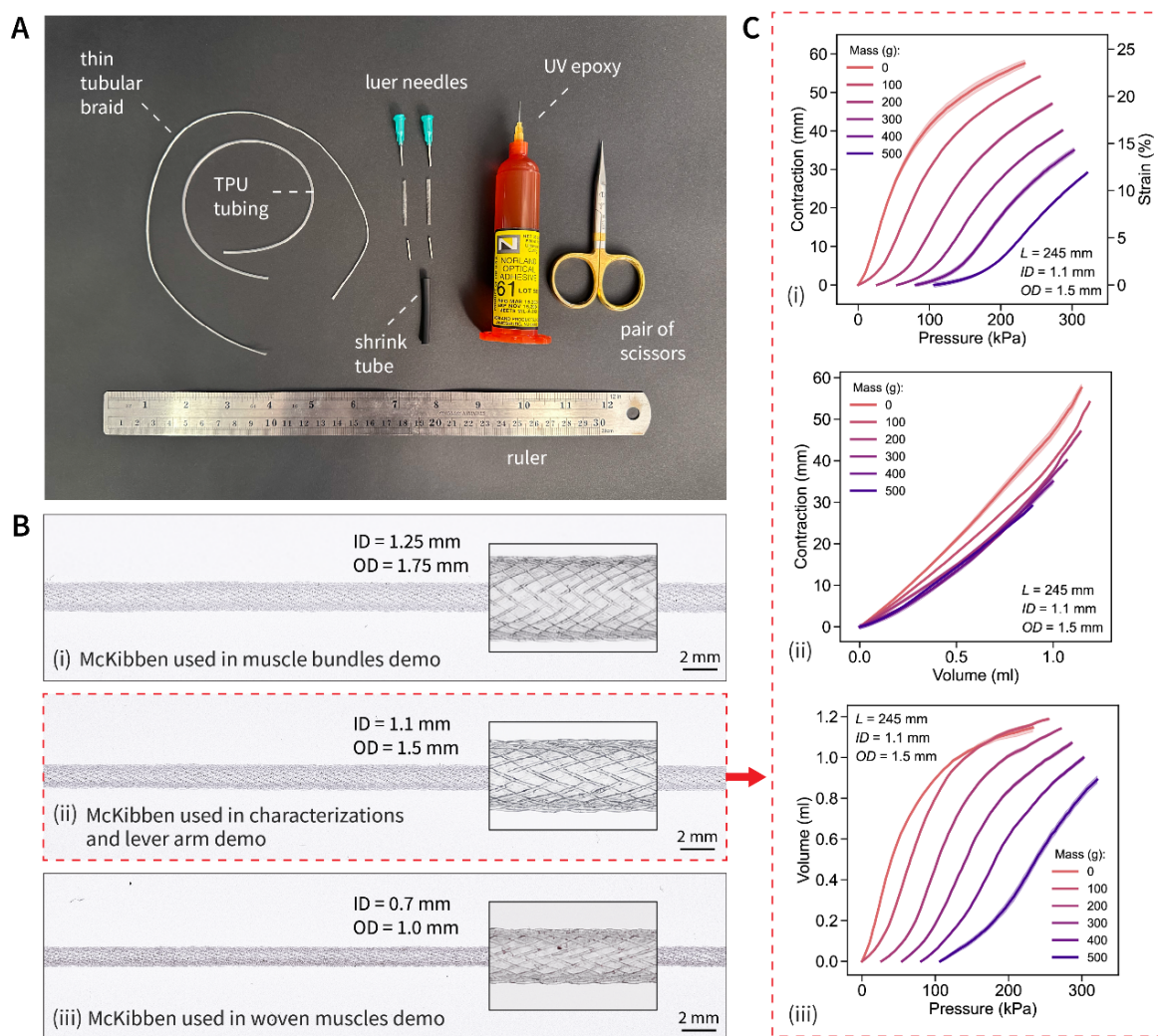


Fig. S4. Thin McKibben actuator fabrication and characterization (A) Bill of materials (BoM) for the manufacture of thin McKibben actuators using thermoplastic polyurethane (TPU, shore 40A) as the core elastomer tubing. The tubular outer braid is made of $n = 24$ nylon monofilaments. (B) Different diameters of McKibben actuators used in the article: (i) woven muscle demo, (ii) all devices used in characterization experiments and lever arm demo, (iii) muscle bundle demo. (C) Thin McKibben actuator response curves ($ID = 1.1$ mm, $OD = 1.5$ mm): (i) contraction versus pressure, (ii) contraction versus volume, (iii) volume versus pressure.

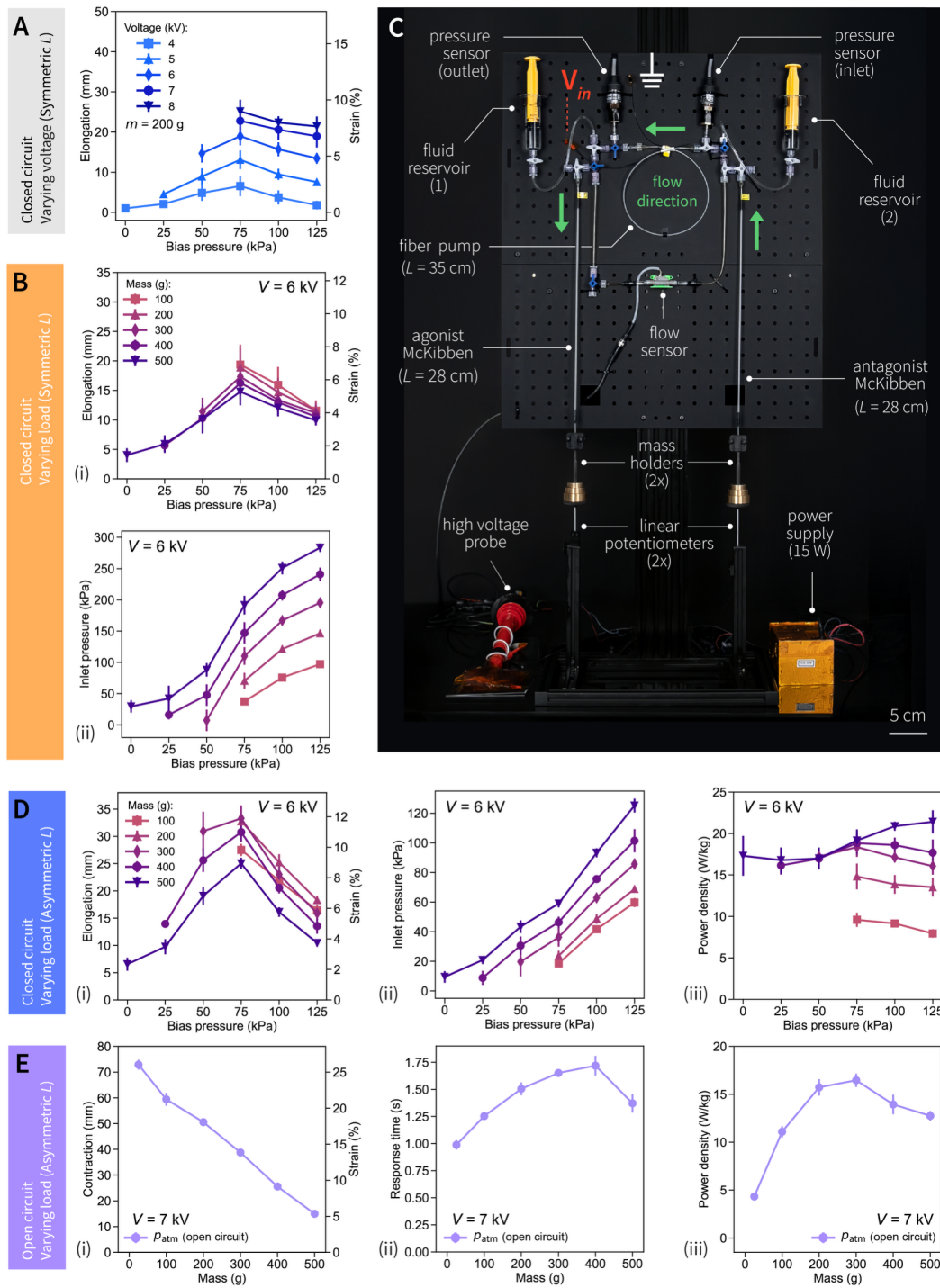


Fig. S5. Full experimental setup. (A) Experimental results of elongation at different bias pressures at voltage steps ranging from 4 to 8 kV, under a fixed symmetric load of 200 g. (B) Elongation (i) and inlet pressure (ii) at different bias pressures, with loads ranging from 100 to 500 g in symmetric loading configuration, at 6 kV. (C) Experimental setup for the characterization of the fiber pumps and muscle pairs. The muscle is activated by a 15-W power supply. During operation, a flow meter measures flow rate, two pressure sensors measure inlet and outlet pressures, two displacement sensors measure actuators' contraction and elongation. The system is filled with dielectric liquid and pre-pressurized with two syringes. (D) Elongation (i), inlet pressure (ii) and power density (iii) at different bias pressures, with loads ranging from 100 to 500 g in asymmetric loading configuration, at 6 kV. (E) Contraction (i), response time (ii) and power density (iii) as a function of the added mass, ranging from preload of 25 g up to 500 g applied mass in asymmetric loading configuration, at 7 kV in open circuit. All values with error bars correspond to mean $\pm$ SD for $n = 3$ devices.

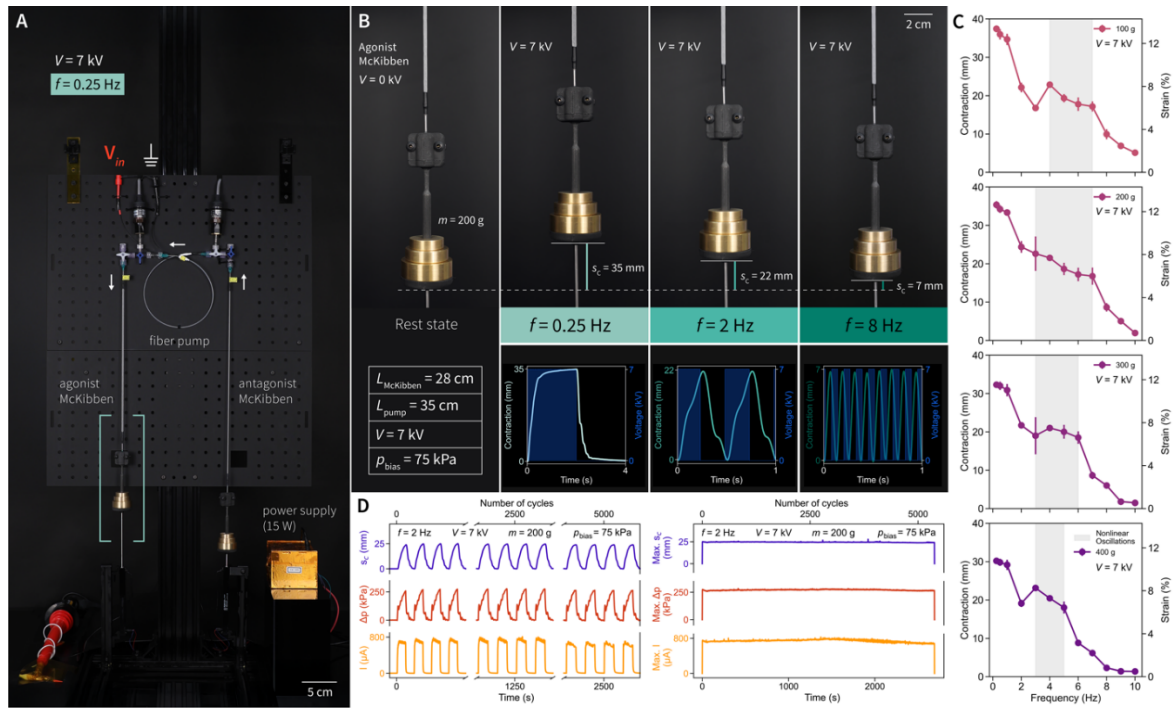


Fig. S6. Dynamic response and cyclic testing of individual muscle pairs. (A) Individual muscle pair experimental setup at actuated state under 200 g load, 7 kV, 0.25 Hz. (B) Zoom-in view of the agonist actuator end point under 75 kPa of bias pressure and 200 g of applied mass at rest state, and when a voltage of 7 kV is applied at frequencies of 0.25, 2 and 8 Hz. (C) Variation in contraction for different frequencies from 0.25 Hz to 10 Hz. Nonlinear oscillations were observed in specific ranges of frequency, highlighted with the grey-shaded area. This area shifts towards the left as the applied mass increases from 100 to 400 g. Values with error bars correspond to mean $\pm$ SD for $n = 3$ devices. (D) Cyclic tests of individual muscle pairs for 5000 cycles at 2 Hz lifting 200 g at 7 kV, with 75 kPa of bias pressure. Contraction (s_c), delta pressure (Δp) and current (I) experienced no significant change from the beginning to the end of the test. Maximum values are evaluated within each cycle. The maximum currents were evaluated after applying a low-pass filter (cutoff frequency: 10 Hz) on the current raw data.

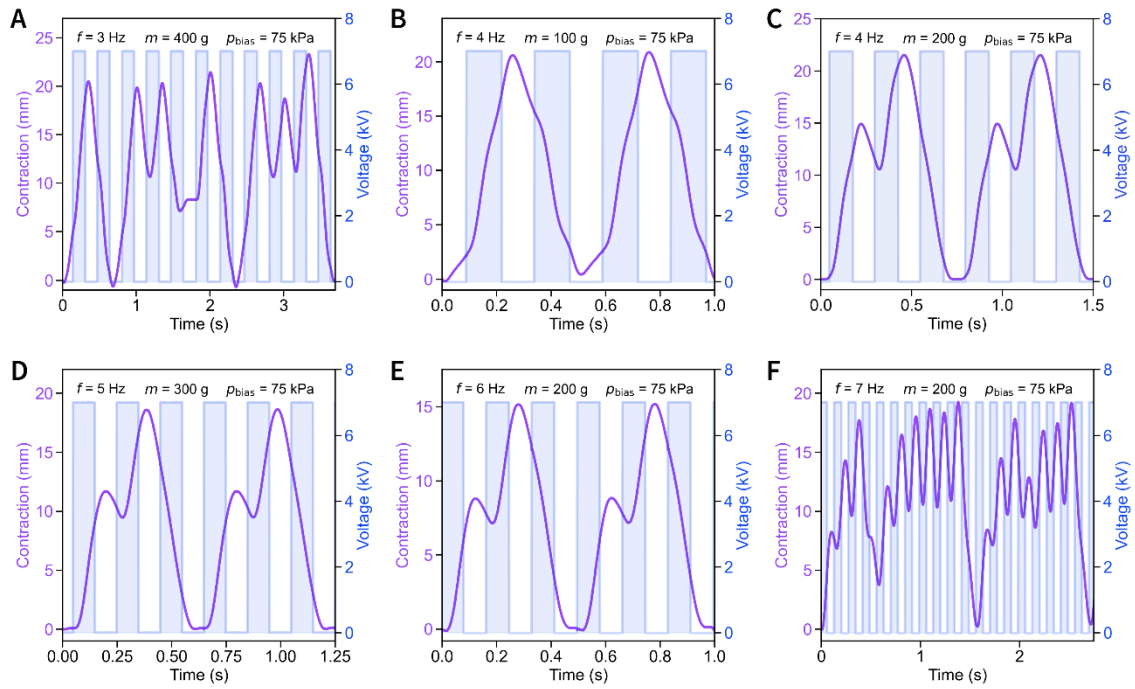


Fig. S7. Non-periodic behavior of the muscle pair during dynamic response experiments. Examples of non-periodic oscillations (dampened periods, double peak periods, chaotic periods) observed at various frequencies (f) and applied values of mass (m). The bias pressure is 75 kPa across all tests. McKibben actuator's oscillations do not follow the oscillation period of the input voltage (7 kV). (A) Chaotic periods at $f = 3$ Hz and $m = 400$ g. (B) Dampened periods at $f = 4$ Hz and $m = 100$ g. (C) Double peak periods at $f = 4$ Hz and $m = 200$ g. (D) Double peak periods at $f = 5$ Hz and $m = 300$ g. (E) Double peak periods at $f = 6$ Hz and $m = 200$ g. (F) Chaotic periods at $f = 7$ Hz and $m = 200$ g.

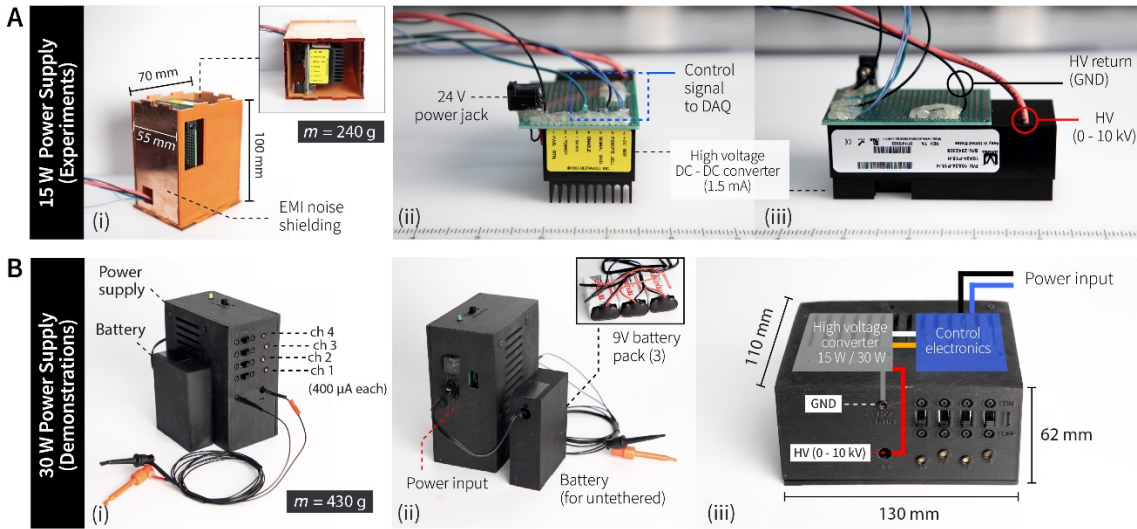


Fig. S8. Power supplies used in the article. (A) Custom 15 W power supply used in all muscle characterization experiments, integrated with a dual data acquisition (DAQ) setup. (i) The power supply is housed in an acrylic enclosure and wrapped with grounded copper shielding to reduce electromagnetic interference (EMI) noise. (ii) The high-voltage DC converter used is Advanced Energy ULTRAVOLT 10A series (15 W maximum power output, 1.5 mA maximum output current), delivering programmable outputs up to 10 kV. The converter is powered by a 24 V input through a power jack. Three control pins interface with the DAQ to modulate the high-voltage output: V_{prog} (input) sets the HV output voltage level, V_{ref} (output) provides the control signal reference, and SGND provides the signal ground. (iii) Shows two high-voltage terminals: HV output (0-10kV) and HV return (GND), which form the actuation circuit. (B) Custom 30-W power supply used in all demonstrations including a control board with four independently addressable HV output channels (up to 400 μA each) (i) and a 9V battery pack for untethered operations (ii). Supply components include a high-voltage DC converter, integrated control electronics and HV output terminals (iii). Dimensions and mass are indicated.

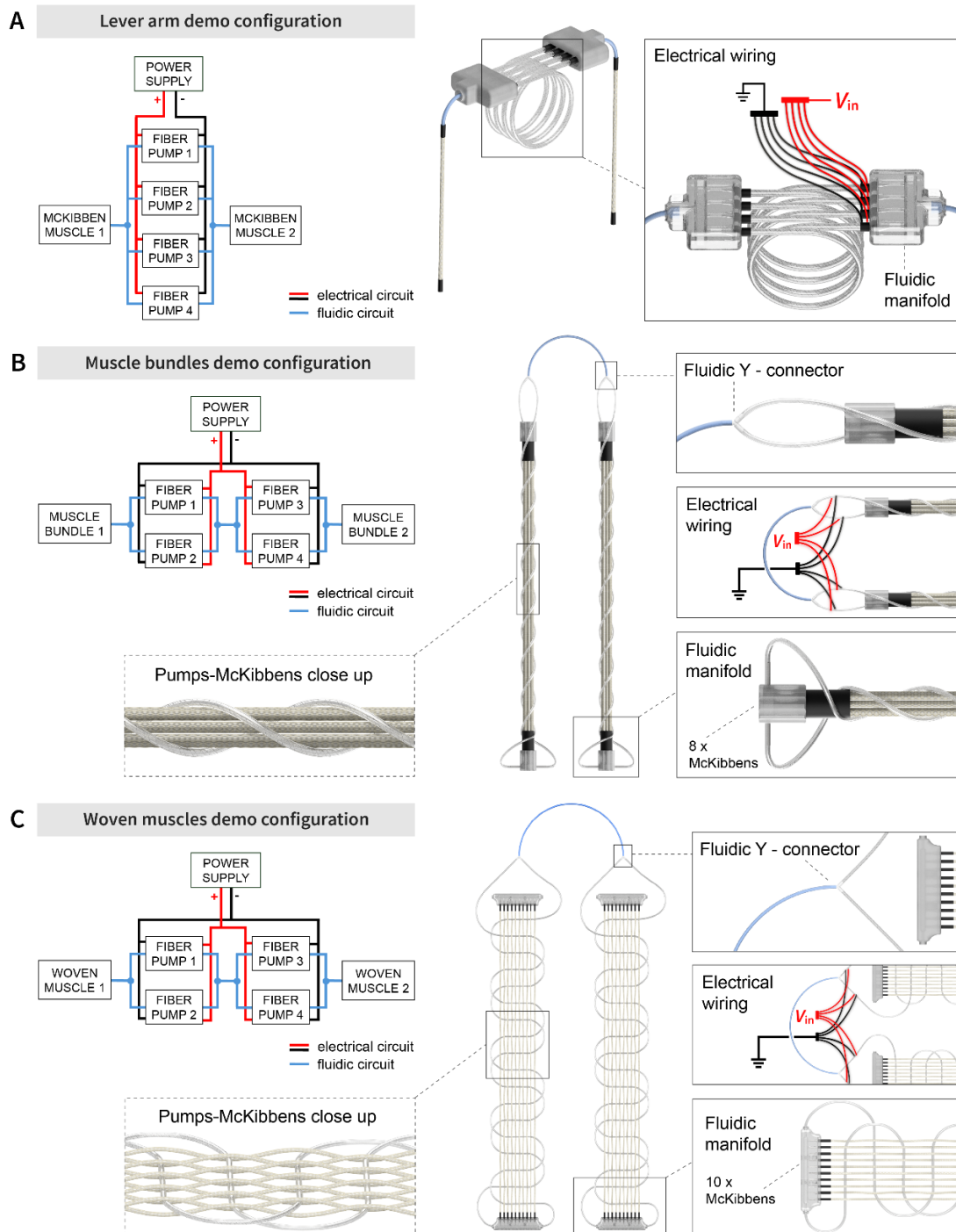


Fig. S9. Schematics for different modular configurations used in the demonstrations. (A) In the lever arm demo, four fiber pumps are fluidically connected in a parallel to drive an antagonistic pair of McKibben actuators. This configuration demonstrates a higher flow rate due to the high pump-muscle ratio, and results in a reduced response time allowing for rapid actuation. (B) In the muscle bundles demo, two antagonistic 8-McKibben actuator bundles are driven by four fiber pumps arranged in parallel-series configuration. A pair of fiber pumps is fluidically connected in parallel using a Y-connector and then connected in series with the second pair of parallel fiber pumps. (C) In the woven muscles demo, two antagonistic 10-McKibben flat actuators are driven by four fiber pumps. A pair of fiber pumps is fluidically connected in parallel using a Y-connector and then connected in series with the second pair of parallel fiber pumps. In all three demonstrations, fiber pumps are fluidically connected in series with the antagonistic McKibben pairs. All four pumps are electrically connected in parallel and driven by a 30-W power supply. The fluidic manifolds are respectively designed and 3D printed.

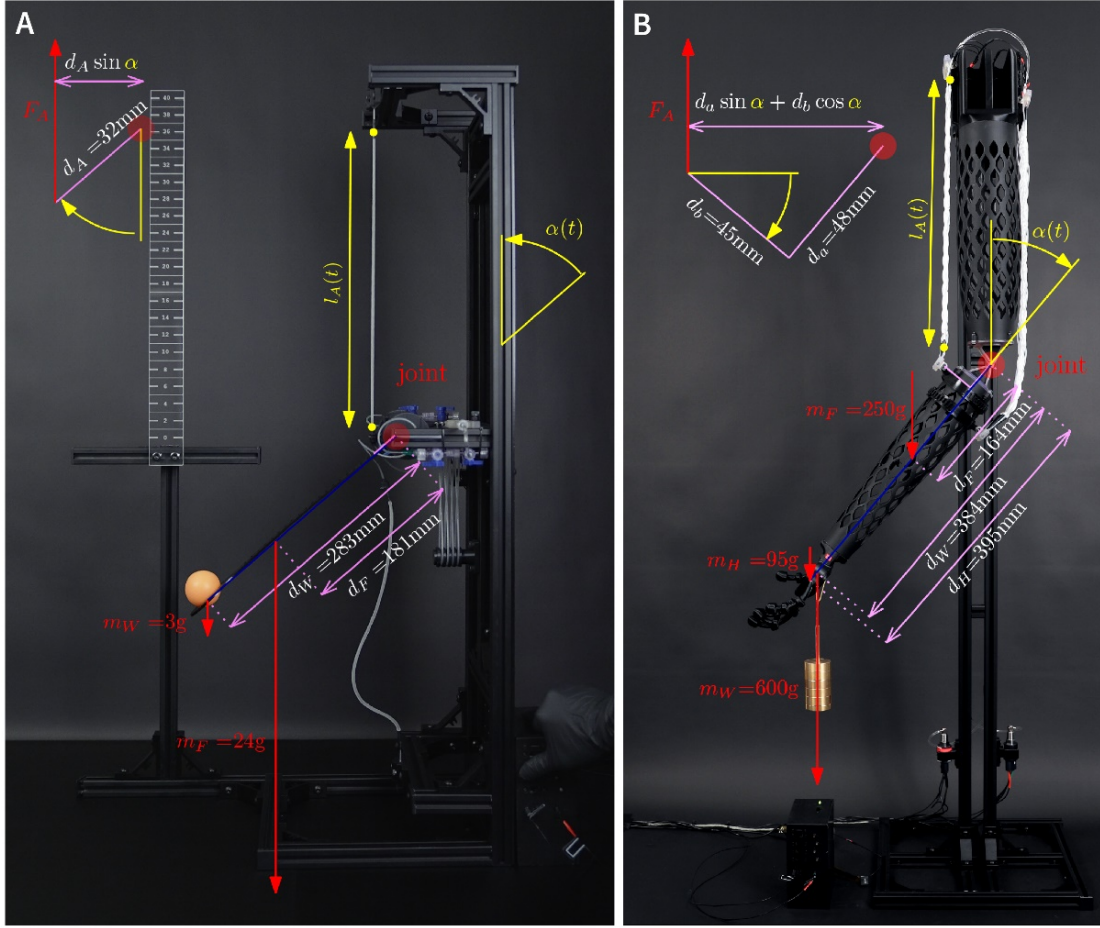


Fig. S10. Force and contraction rate of woven sheet muscle and lever arm. The angles and the contracting muscle lengths in time are obtained by tracking them using a video analysis software. The contraction rates reported in Fig. 1E (iv), are the ratio between the maximum contraction and the response time at 90% contraction. The contracting muscle forces in Fig. 1 are computed by balancing the torques around the joint, neglecting inertia, and the extending muscle contributions. The distances between the joint and components' centers of mass are computed from the respective CAD files for each demo, and the weights of physical components are measured manually. (A) The sampling rate is 600 Hz. The force is computed at the time when the lever hits the stopper, i.e. the ball leaves the holder. (B) The tracking is conducted at a sampling rate of 30 Hz. The force is computed at maximum arm flexion. The reported values are averaged over 10 actuation/lifting cycles.

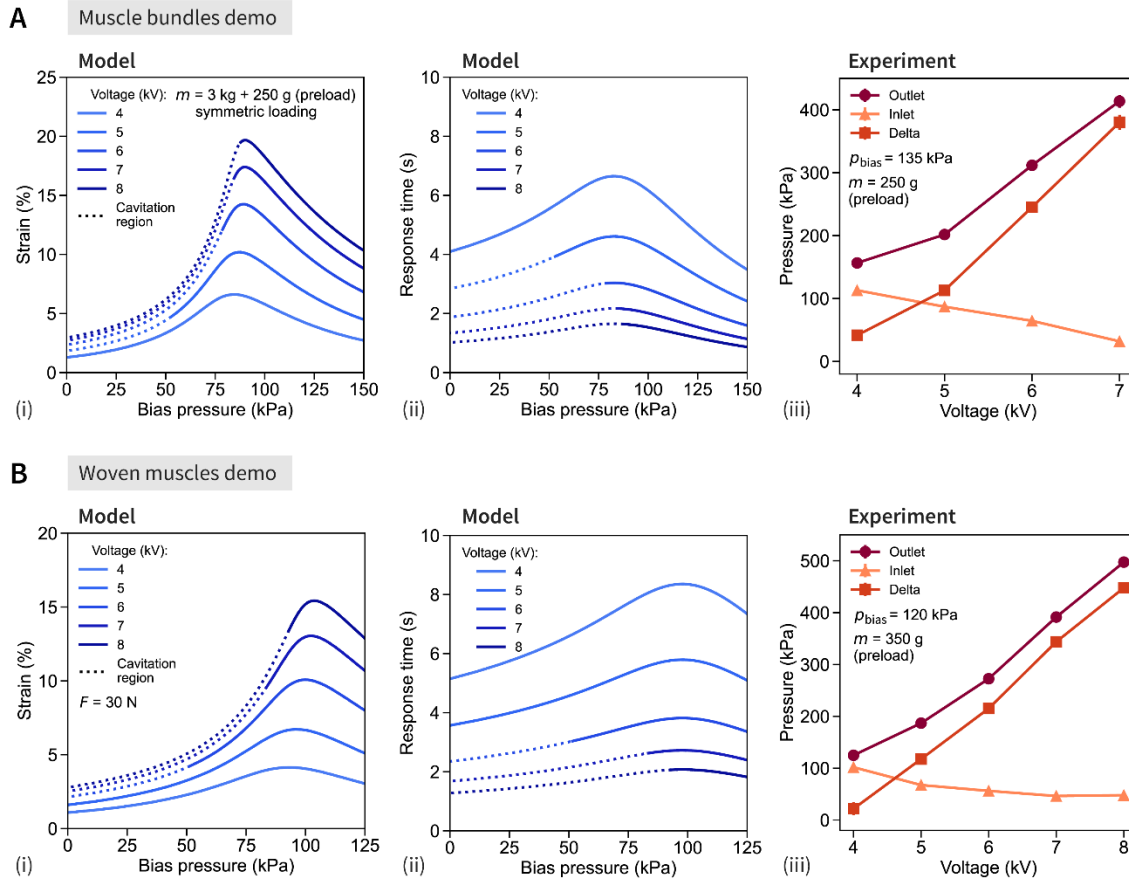


Fig. S11. Model predictions and pressure measurements for woven muscles and muscle bundles demonstrations. (A) Model prediction for muscle bundles showing optimal bias pressure for strain (i), and response time (ii). Inlet, outlet and delta pressure measurements in the varying voltage experiment of the muscle bundles demo (iii). Inlet pressure at 7 kV is nearly 30 kPa. At 8 kV, this value is expected to drop below the threshold for safe operation ($< 0 \text{ kPa}$) according to the pump's voltage-pressure relationship shown in Fig. 2B (ii). Therefore, the muscle bundles were not tested beyond 7 kV. (B) Model prediction for woven muscles showing optimal bias pressure for strain (i), and response time (ii). Inlet, outlet and delta pressure measurements in the varying voltage experiment of the woven muscles demo. Inlet pressure at 8 kV is nearly 30 kPa and the outlet pressure is 500 kPa (iii). At higher voltage values, the inlet pressure is expected to drop below the threshold for safe operation ($< 0 \text{ kPa}$), while the outlet pressure would approach McKibben's burst pressure (550 kPa). Therefore, the woven muscles were not tested at higher voltage values.

SUPPLEMENTARY TABLES

	Reference	Length (mm)	Stroke (mm)	Strain (%)	Device mass (g)	Response time (s)	Recovery time (s)	Force (N)	Strain rate (%/s)	Specific force (N/kg)	Power density (W/kg)
Mammalian skeletal muscles	(59)	-	-	20.0 (typ) 40.0 (max)	-	-	-	-	500 (max)	-	50.0 (avg) 200 (max)
Electrofluidic muscles											
Muscle pair (optimal p_{bias})	This work	280	40.2	14.4	4.30 ^a	0.90	0.35	4.90	16.0	1140	51.0
Muscle pair (min. t)		280	27.0	9.64	4.80 ^a	0.28	0.22	1.96	34.4	409	-
Lever arm		340	24.7	7.30	8.00 ^a	0.13	-	1.35	55.8	168	32.0
Bundle		280	30.0	10.7	22.0 ^a	3.55	1.00	41.7	3.00	1900	16.0
Open system (max. strain)		280	72.8	26.0	4.20 ^a	1.00	0.81	0.25	26.0	58.4	4.25
Open system (max. P_d)		280	38.7	13.8	4.20 ^a	1.65	0.40	2.94	8.36	700	16.4
DEA fibers											
Actuator c	(28)	25.0	1.00	4.00	0.003	2.00	2.00	0.006	2.00	2000	0.98 ^b
Actuator c*	(28)	-	-	9.00	-	0.10	0.10	0.001	90.0	-	-
Bundle	(28)	25.0	0.80	3.20	0.012	-	-	0.02	-	1800	-
Long artificial muscle	(29)	160	5.00	3.13 ^c	8.50	0.17 ^c	0.17 ^c	0.10 ^d	18.8	11.5	0.35 ^b
Textuators											
6T yarns*	(60)	20.0	0.05	0.25	-	600	290	0.06	0.0004	-	-
12 T yarns*	(60)	20.0	0.06	0.32	-	620	510	0.10	0.0005	-	-
Knitted S-yarns*	(60)	20.0	0.6	3	-	550	350	0.01	0.01	-	-
Electrochemical muscles											
CNT - Hybrid yarn*	(36)	15.0	1.22	8.10	0.0003 ^e	3.00 ^f	-	0.04 ^g	2.70	131000	54.3 ^b
Electrothermal muscles											
LCE – fiber †	(34)	128 ^h	55.0	43.0	-	20.0 ⁱ	-	0.08	2.15	-	-
LCE – DTWA double fiber*	(34)	79.0 ^h	29.4	37.2	-	6.00	10.0	2.50	6.20	-	-
CuNF-SMAc – natural cooling †	(30)	-	-	50.0	0.06	2.00	15.0	2.50	25.0	40900	71 ^j
CF@PDMS †	(35)	-	-	16.0	-	10.0	10.0	0.27	1.60	-	-

Table S1. Comparison table for electrically-driven artificial fiber muscles. Mammalian skeletal muscles are included as a reference and benchmark.

* It requires the system to be in a liquid bath (water, electrolyte, etc.).

† It reaches high temperatures during actuation ((34) ~200°C, (30) ~50 °C, (35) ~150 °C)

^a Device mass includes the total fluid mass in bias pressure (additional fluid mass) condition.

- ^b Power density not reported. An estimate has been calculated from the reported work capacity (W_c) and response time (t) using $P_d = \frac{W_c}{t}$. If work capacity is not reported, the estimate is calculated using $P_d = \frac{F s}{m t}$, where F is the force, s is the stroke and m is the mass of the system.
- ^c Calculated considering the non-resonant modulated amplitude actuation case of 3 Hz – 50% duty cycle (Figure 4C in (29)). At resonance (10 Hz), they report a power density of 7.8 W/kg and a maximum strain of 8.9%.
- ^d Force not reported. An estimate has been calculated based on the mass shown in Figure 4A and movie S6 in (29).
- ^e Mass not reported. An estimate has been calculated from the reported work capacity and the estimated force and response time.
- ^f Response time estimated from the reported strain (st) and strain rate (st_r) in Table S1 in (36) using $t = \frac{st_r}{st}$.
- ^g Force calculated from the reported tensile stress (σ) and fiber diameter (d) using $F = \sigma \left(\frac{d}{2}\right)^2 \pi$.
- ^h Length not reported. An estimate has been calculated from the figures and the reported scale bar. It refers to the length of the fiber under the applied load.
- ⁱ Response time not reported. An estimate has been calculated considering the time to reach the 90% of the maximum strain (Figure 2C in (34)).
- ^j Reported power density uses a different formula. An estimate was calculated using our formula, considering the reported maximum actuation speed (Fig. 4E in (30)), which might overestimate the average power density.

	No liquid	With liquid					
		Applied bias pressure (kPa)					
		0	25	50	75	100	125
Muscle pair mass (g)	1.8	3.3	3.6	4.0	4.3	4.6	4.8

Table S2. Mass of the individual fiber muscle pair when unfilled or filled at different bias pressure values. The muscle pair consists of two McKibben actuators ($ID = 1$ mm, $OD = 1.5$ mm, $L = 280$ mm), one fiber pump ($ID = 1.1$ mm, $OD = 1.6$ mm, $L = 350$ mm), two stainless steel blunt needles, and two 10 cm-long HV leads. As expected, system mass increases as the applied bias pressure increases.

LEGENDS FOR MOVIES

Movie 1. Working principle and demonstration of self-contained electrofluidic fiber muscles

Movie S1. Electrofluidic fiber muscle pair in open vs. closed fluidic circuit configurations

Movie S2. Determining cavitation thresholds for electrofluidic fiber muscles

Movie S3. Performance characterization of electrofluidic fiber muscle pair

Movie S4. Dynamic response of electrofluidic fiber muscles

Movie S5. Assembly of lever arm demonstration using four fiber pumps connected in parallel

Movie S6. Demonstrating rapid actuation: a lever arm driven by four fiber pumps connected in parallel

Movie S7. Fabrication of electrofluidic muscle bundles using four fiber pumps connected in parallel – series

Movie S8. Demonstrating high force: muscle bundle pair lifting > 4 kg using four fiber pumps connected in parallel – series

Movie S9. Fabrication of electrofluidic woven muscles using four fiber pumps connected in parallel – series

Movie S10. Demonstrating large range of motion: model arm driven by a woven muscle pair using four fiber pumps connected in parallel – series

Movie S11. Untethered operation of electrofluidic woven muscle